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Analysis and Classification of Texts Generated by Deceptive Queries to Language Models

A Machine Learning Based Approach

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Abstract

In the context of modern artificial intelligence applications, the need for accurate and reliable tools to classify the answers generated by Large Language Models (LLMs) represents a crucial challenge, especially when these models are subjected to prompts aimed at compromising their alignment (jailbreak). This work aims to fill this gap through the design and development of an evaluation system based on machine learning techniques.

Starting from an in-depth analysis of the existing approaches, we identified transformer models as a solid starting point for the implementation of a fine-tuned classifier able to distinguish between “jailbreak” answers and answers compliant with the expected behaviour. The training process followed an incremental workflow, characterised by periodic reviews and continuous feedback, in order to maximise the robustness and the generalisation ability of the model.

To complement the classification engine, a clustering stage based on the K-means algorithm was integrated, aimed at exploring latent structures in the data and at providing a mechanism for grouping the answers according to their semantic similarity. The software architecture was organised into distinct modules, which clearly separate the core devoted to training and saving the fine-tuned model from the analysis and visualisation components, thus ensuring ease of extension and maintenance.

The implementation relies on open-source Python libraries and is based on containerised workflows in order to guarantee reproducibility and portability. The validation tests that were carried out showed that the system offers superior performance with respect to the non-fine-tuned version of the reference model, with percentage increases in accuracy that confirm its effectiveness.

Contents

1	Introduction	8
1.1	Research context and motivations	8
1.2	Relevance and impact of the research	9
1.3	System architecture	9
1.4	Methodology and implementation	10
1.5	Applications and implications	10
1.6	Structure of the document	11
2	Theoretical foundations	12
2.1	Large Language Models (LLMs)	12
2.2	Foundations of machine learning	13
2.2.1	Learning paradigms	13
2.2.2	Neural networks and deep learning	14
2.2.3	The training and validation process	14
2.2.4	Transfer learning and fine-tuning	15
2.2.5	Regularisation and generalisation	15
2.3	The Transformer architecture	16
2.4	The BERT model: pre-training and fine-tuning	16
2.4.1	Extraction of semantic embeddings from BERT	17
2.5	Model alignment and jailbreaking	17
2.6	Analysis through clustering with K-means	17
2.6.1	Evaluation of clustering quality	18
2.7	Dimensionality reduction techniques for visualisation	18
3	State of the art	20
3.1	Taxonomy of jailbreaking attacks	20
3.2	Defence mechanisms and evaluation platforms	20
3.3	Automatic evaluation of LLM answers	21
3.3.1	Evaluation based on traditional metrics	21
3.3.2	Evaluation through LLMs (LLM-as-a-Judge)	21
3.3.3	Evaluation through specific classifiers	21
3.4	Limitations of the existing approaches and positioning of the thesis	22

4	Methodology	24
4.1	Introduction	24
4.2	Overview of the methodological approach	24
4.2.1	Evolution of the research framework	24
4.3	Experimental setup and models used	29
4.3.1	Dual-model comparative architecture	29
4.3.2	Evaluation dataset and preprocessing	29
4.3.3	Experimental configuration parameters	30
4.4	Extraction and representation of the embeddings	30
4.4.1	Detailed extraction process	30
4.4.2	Quality and properties of the embeddings	31
4.5	Clustering algorithms and optimisation	32
4.5.1	K-Means: foundations and implementation	32
4.5.2	Optimisation metrics	32
4.5.3	Validation and quality metrics	33
4.6	Visualisation and interpretation	33
4.6.1	Dimensionality reduction techniques	33
4.6.2	Multi-level visualisations	34
4.7	Validation and robustness	34
4.7.1	Validation protocol	34
4.7.2	Handling of uncertainty	34
4.8	Final implementation pipeline	35
4.8.1	System architecture	35
4.8.2	Data management and preprocessing	35
4.9	Methodological conclusions	35
5	Presentation and analysis of the results	37
5.1	Introduction	37
5.2	Results of version 1.0: simplified two-dimensional approach	37
5.2.1	Methodological configuration and initial results	37
5.2.2	Limitations of the two-dimensional approach	39
5.3	Results of version 2.0: high-dimensional embeddings with fixed clustering	39
5.3.1	Transition to complete embeddings	39
5.3.2	Dataset and original composition	39
5.3.3	Clustering results on the complete dataset	39
5.3.4	Analysis of accuracy and correspondence with the ground truth .	40
5.3.5	Detailed analysis with confusion matrices	41
5.3.6	Clustering quality metrics	41
5.3.7	Validation through dimensional visualisations	42
5.3.8	Interpretation of the differences	44

5.3.9	Implications for the effectiveness of the system	44
5.4	Results of version 3.0: adaptive clustering with automatic determination of the optimal number of clusters	45
5.4.1	Transition to adaptive clustering	45
5.4.2	Dataset and optimisation methodology	45
5.4.3	Results of the automatic clustering	46
5.4.4	Analysis of the optimisation metrics	46
5.4.5	Analysis of accuracy and correspondence with the ground truth .	47
5.4.6	Detailed analysis with confusion matrices	48
5.4.7	Validation through dimensional visualisations	49
5.4.8	Advanced clustering quality metrics	51
5.4.9	Semantic interpretation of the tri-cluster structure	52
5.4.10	Performance comparison: fixed vs adaptive clustering	52
5.4.11	Implications for the architecture of the system	53
5.4.12	Conclusions of version 3.0	53
5.5	Results of version 4.0: final hierarchical clustering system	54
5.5.1	Transition to multi-level hierarchical clustering	54
5.5.2	Dataset and hierarchical architecture	54
5.5.3	Results of the hierarchical clustering	55
5.5.4	Analysis of the hierarchical optimisation metrics	55
5.5.5	Validation through multi-dimensional visualisations	57
5.5.6	Quantitative analysis of the hierarchical quality	58
5.5.7	Implications of the hierarchical system	59
6	Conclusions	62
6.1	Main results and findings	62
6.2	Alignment with the research objectives	63
6.3	Relevance and practical applications	63
6.4	Critical comparison between the final approaches	64
6.5	Limitations and methodological challenges	64
6.6	Contributions to the existing body of knowledge	65
6.7	Future directions and prospective developments	65
6.8	Overall significance and closing remarks	66
	Bibliography	67
	Acknowledgements	69

List of Tables

5.1	Summary results of version 1.0 on a sample of 20 answers	38
5.2	Comparison of the cluster distributions: BERT Fine-tuned vs BERT Base	40
5.3	Accuracy of the clustering with respect to the original labels	40
5.4	Confusion matrix - BERT Fine-tuned	41
5.5	Confusion matrix - BERT Base	41
5.6	Separability and compactness metrics of the clusters	42
5.7	Cluster distribution: automatic vs fixed clustering	46
5.8	Detailed optimisation metrics for the range k=2-10	47
5.9	Accuracy of the automatic clustering with respect to the original labels .	47
5.10	Confusion matrix - BERT Fine-tuned (3 clusters)	48
5.11	Confusion matrix - BERT Base (2 clusters)	48
5.12	Geometric quality metrics of the clusters - comparison of the configurations	51
5.13	Inter-cluster distances and intra-cluster compactness	51
5.14	Comparative performance: fixed vs adaptive approaches	52
5.15	Hierarchical distribution of the clusters: comparison of BERT Fine-tuned vs BERT Base	55
5.16	Detailed optimisation metrics for the hierarchical clustering	56
5.17	Comparative hierarchical quality metrics	58

List of Figures

1.1	Complete pipeline of the evaluation system	10
5.1	Two-dimensional distribution of the results of version 1.0. The X axis represents the binary classification (-1 = no-jailbreak, +1 = jailbreak), while the Y axis shows the confidence of the model.	38
5.2	PCA projection with BERT Fine-tuned. Extremely sharp separation with well-defined geometries.	42
5.3	PCA projection with BERT Base. Less sharp separation with greater overlap.	42
5.4	t-SNE projection with BERT Fine-tuned. Extremely compact and well-separated clusters.	43
5.5	t-SNE projection with BERT Base. Greater dispersion and internal fragmentation.	43
5.6	Elbow and Silhouette analysis - BERT Fine-tuned. The optimal point at k=3 is clearly identifiable, with a maximum Silhouette Score of 0.61. . .	46
5.7	Elbow and Silhouette analysis - BERT Base. Convergence towards k=2 with a maximum Silhouette Score limited to 0.20.	46
5.8	PCA projection - BERT Fine-tuned (3 clusters). Extremely sharp linear separation with three geometrically distinct and compact regions.	49
5.9	PCA projection - BERT Base (2 clusters). Binary separation with significant overlap and structural dispersion.	49
5.10	t-SNE projection - BERT Fine-tuned (3 clusters). Extremely compact clusters with optimal inter-cluster separation and an evident hierarchical structure.	50
5.11	t-SNE projection - BERT Base (2 clusters). Binary configuration with significant intra-cluster dispersion and blurred boundaries.	50
5.12	UMAP projection - BERT Fine-tuned (3 clusters). Extremely well-defined topology with dense clusters and preserved local connectivity.	50
5.13	UMAP projection - BERT Base (2 clusters). Binary structure with a less defined topology and greater fragmentation.	50
5.14	Elbow & Silhouette analysis - BERT Base, Macro-cluster 0 (no-jailbreak). Optimum at k=3 with a Silhouette Score of 0.332.	56

5.15	Elbow & Silhouette analysis - BERT Fine-tuned, Macro-cluster 0 (no-jailbreak). Optimum at k=2 with a Silhouette Score of 0.438.	56
5.16	Elbow & Silhouette analysis - BERT Base, Macro-cluster 1 (jailbreak). Optimum at k=2 with a Silhouette Score of 0.128.	56
5.17	Elbow & Silhouette analysis - BERT Fine-tuned, Macro-cluster 1 (jailbreak). Optimum at k=2 with a Silhouette Score of 0.655.	56
5.18	PCA projection - Macro-clusters BERT Base. Limited separation with significant overlap between the main categories.	57
5.19	PCA projection - Macro-clusters BERT Fine-tuned. Extremely sharp separation with geometrically distinct and compact regions.	57
5.20	t-SNE projection - Sub-clusters BERT Base. Fragmented structure with poorly cohesive subcategories and indistinct boundaries.	58
5.21	t-SNE projection - Sub-clusters BERT Fine-tuned. Highly cohesive sub-structures with optimal inter-cluster separation.	58
5.22	t-SNE 2D - Macro-cluster view BERT Base (k=2)	60
5.23	t-SNE 2D - Macro-cluster view BERT Fine-tuned (k=2)	60
5.24	UMAP 2D - Macro-cluster view BERT Base (k=2)	60
5.25	UMAP 2D - Macro-cluster view BERT Fine-tuned (k=2)	60
5.26	PCA 2D - All the labelled subclusters, BERT Base	60
5.27	PCA 2D - All the labelled subclusters, BERT Fine-tuned	60
5.28	UMAP 2D - All the labelled subclusters, BERT Base	61
5.29	UMAP 2D - All the labelled subclusters, BERT Fine-tuned	61

CHAPTER 1

Introduction

Artificial intelligence has undergone a profound transformation in recent years, mainly thanks to the emergence of Large Language Models (LLMs). These systems, based on complex neural architectures and trained on vast textual corpora, have shown extraordinary capabilities in understanding and generating natural language, revolutionising fields ranging from virtual assistance to the automatic production of content, from machine translation to advanced textual analysis.

However, alongside the remarkable potential of these models, significant critical issues have emerged that raise fundamental questions about their reliability and safety. One of the most relevant challenges concerns the vulnerability of these systems to deceptive queries, commonly referred to as *jailbreak prompts*, which can induce the models to produce inappropriate, misleading, illegal or potentially dangerous answers. This phenomenon represents a violation of the alignment of the model, that is, of its ability to adhere to the ethical principles and to the guidelines established during the training phase.

The problem of the alignment of language models constitutes one of the most critical research topics in the field of artificial intelligence. The growing integration of these systems into critical applications makes it necessary to develop effective methodologies to assess and guarantee their robustness against attempts at manipulation and improper use of these models. The complexity of such a challenge is amplified by the evolving nature of the attack techniques, which can be developed and refined through sophisticated attack tools, including evolutionary algorithms able to automatically generate and evolve prompts that are increasingly effective at compromising the alignment of the model.

1.1 Research context and motivations

The present work is placed within this research landscape, and specifically addresses the problem of the automatic evaluation of the answers generated by language models in response to potentially compromising queries. The study builds on the results previously obtained by Brighenti Stefano, who fine-tuned a (base) variant of the BERT model

(Bidirectional Encoder Representations) specifically for the binary classification of LLM answers as *jailbreak* or *non-jailbreak*. Starting from this consolidated methodological basis, the present research extends the analysis through the application of unsupervised clustering techniques in order to explore hidden patterns in the vector representations of the classified answers.

The primary objective of this study is to develop and validate an integrated approach that combines supervised classification based on BERT with clustering techniques, implemented through the K-means algorithm, in order to analyse the intrinsic characteristics of the answers generated by language models when subjected to prompts produced by evolutionary systems. This methodological approach makes it possible to gain a deeper understanding of the ways in which language models react to attempts to compromise their alignment, providing valuable insights for the development of more sophisticated evaluation systems.

1.2 Relevance and impact of the research

The relevance of this investigation can be appreciated on multiple levels. From a theoretical point of view, the work carried out aims to contribute to the advancement of knowledge in the field of the security of language models, a rapidly expanding research area that involves academic institutions and research centres worldwide. From a practical point of view, the results obtained provide a baseline methodology for the design and construction of concrete tools for the automatic evaluation of the robustness of language models, which are essential elements in guaranteeing a safe and reliable deployment of these systems in real application contexts.

The present work is part of a broader research project carried out at the Department of Informatics, Systems and Communication (DISCo) of the University of Milano-Bicocca, under the supervision of Prof. Claudio Ferretti and Prof. Martina Saletta. This project investigates the mechanisms by which the alignment of language models is broken, through the use of evolutionary algorithms for the generation of attack prompts.

Within this research pipeline, the specific contribution of this dissertation concerns the answer evaluation component, a crucial element that allows the evolutionary algorithms to iteratively optimise their ability to generate effective prompts. Without an accurate and automated evaluation system, it would be impossible to provide the feedback needed to drive the evolution of the prompts towards configurations that are increasingly effective at compromising the alignment of the target model.

1.3 System architecture

In order to provide an overview of the research framework in which this work is placed, Figure 1.1 illustrates the complete pipeline of the system created by Prof. Ferretti and

Prof. Saletta. As highlighted in the diagram, the process is articulated into four main components: the evolutionary generation of the attack prompts, the querying of the target LLM, the extraction and preprocessing of the answers, and finally the automatic evaluation through classification and clustering. It is precisely this last component, represented by the fourth element of the pipeline, that constitutes the specific focus of this dissertation, where the answer analysis methodologies based on BERT and K-means clustering are implemented and validated.

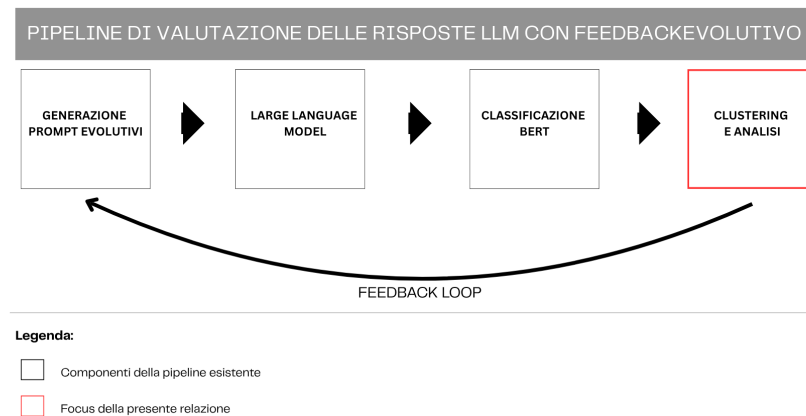


Figure 1.1: Complete pipeline of the evaluation system

1.4 Methodology and implementation

The methodology adopted involves the use of the Google Colab environment for the implementation and the execution of the experimental code, guaranteeing the reproducibility and the accessibility of the results. The fine-tuned BERT model is imported from the Hugging Face platform, ensuring the use of architectures that are standardised and validated by the scientific community. The experimental approach integrates supervised classification with the analysis of the vector representations obtained through the embedding of the answers into the latent space of the BERT model, thus allowing the application of clustering techniques to identify structural patterns in the analysed answers.

1.5 Applications and implications

The results of this research have implications for several application areas. In the field of computer security, they contribute to the development of more effective detection systems for identifying attempts to manipulate language models. In the field of the development

of artificial intelligence systems, they provide methodologies for the continuous evaluation of robustness during the development and deployment phases. Moreover, the insights obtained can offer useful hints for the development of more robust alignment techniques, contributing to the creation of language models that are intrinsically more resistant to attempts at compromise.

1.6 Structure of the document

The structure of this document is organised so as to provide a complete and systematic presentation of the work carried out. Chapter 2 presents the theoretical notions needed to understand the technical foundations of the present study, including the citations to papers and scientific research relevant to the topics addressed. Chapter 3 provides a review of the state of the art, analysing the most significant contributions in the scientific literature on language models, jailbreaking techniques, and classification/clustering methods applied to textual analysis. Chapter 4 describes in detail the methodologies adopted, including the description of the dataset used, the detailed explanation of the techniques for generating the vector representations of the answers by means of the BERT model, the implementation of the K-means algorithm, and the evaluation metrics that were selected. Chapter 5 presents the experimental results obtained, providing a quantitative and qualitative analysis of the patterns identified through clustering and discussing the implications of the results in the context of the robustness of language models. Chapter 6 summarises the main contributions of the research, discusses the limitations of the work and outlines future directions for the extension and the deepening of this line of research. The document ends with the bibliography and the acknowledgements.

CHAPTER 2

Theoretical foundations

In order to fully understand the context and the methodology of this dissertation, it becomes essential to outline the technological and conceptual pillars that constitute the theoretical foundations of this study. This chapter provides an examination of the theoretical concepts that are fundamental for understanding what lies at the basis of the tools used in the present study. Such tools include: Large Language Models (LLMs), the Transformer architecture that made their development possible, and the techniques employed for the analysis. In particular, the BERT model, the ways of extracting semantic representations (embeddings), clustering techniques such as K-means and dimensionality reduction algorithms for visualisation will be examined in depth.

2.1 Large Language Models (LLMs)

Large Language Models (LLMs) represent one of the most significant advances in the field of artificial intelligence. They are computational models based on deep neural networks, trained on textual corpora of massive size, often containing hundreds of billions of words coming from the Internet and from digitised archives. Their primary function is to model the probability distribution of natural language. In simpler terms, an LLM learns to predict the next word in a given sequence, an apparently simple ability which, when scaled sufficiently, gives rise to extraordinary emergent skills such as translation, summarisation, question answering and code generation.

These models are characterised by a number of parameters that can range from a few billion to more than one trillion. The vastness of these parameters, combined with the complexity of the underlying neural architecture and with the diversity of the training data, allows LLMs to capture complex syntactic, semantic and contextual nuances of human language.

2.2 Foundations of machine learning

Before examining in depth specific architectures such as BERT and the Transformers, it is essential to understand the principles of machine learning that constitute their theoretical and methodological basis. Machine learning represents a computational paradigm in which algorithms learn patterns from data without being explicitly programmed for every single case [1]. This ability to generalise from examples constitutes the conceptual core that makes the extraordinary performance of contemporary language models possible.

2.2.1 Learning paradigms

Machine learning is articulated into three main learning paradigms, each characterised by different types of input data and by specific objectives. Understanding these paradigms is essential in order to grasp how the different components of this dissertation are integrated into a coherent methodological framework.

Supervised learning represents the paradigm in which the algorithm learns from a training dataset that contains both the inputs and the desired outputs, called labels. The objective is to build a function that accurately maps the inputs to the outputs, allowing the model to make predictions on new data never seen during training. This paradigm includes two main subcategories: classification, where membership in discrete categories is predicted, and regression, where continuous values are predicted. In the context of this dissertation, the fine-tuning of BERT for jailbreak/non-jailbreak classification represents a paradigmatic example of supervised learning [2].

Unsupervised learning constitutes a fundamentally different approach, in which the algorithm must discover hidden patterns in data that do not contain explicit labels. This paradigm is particularly powerful for the exploration of latent structures in the data, making it possible to identify natural groupings, to reduce dimensionality or to discover compact representations of the information. Clustering techniques such as K-means, used in this dissertation to analyse the structure of the embeddings, belong to this category and represent a fundamental tool for understanding the semantic organisation of the representation space [3].

Self-supervised learning emerges as a hybrid paradigm that is particularly relevant for language models. In this approach, the labels are created automatically from the data themselves, exploiting the intrinsic structure of the information in order to define learning tasks. The pre-training of BERT through the Masked Language Model and Next Sentence Prediction tasks perfectly exemplifies this methodology: the model learns to predict masked words or relations between sentences by using the natural context of the text, acquiring a deep linguistic understanding without the need for expensive and labour-intensive human annotations [4].

2.2.2 Neural networks and deep learning

The models discussed in this dissertation are based on artificial neural networks, computational systems inspired by the functioning of biological neurons but implemented through precise and optimisable mathematical operations. A neural network is composed of layers of computational units called artificial neurons, each of which receives multiple inputs, combines them through a weighted sum, and applies a non-linear activation function in order to produce an output. This non-linearity is crucial because it allows the network to learn complex relations that could not be captured by linear models [5].

Deep learning represents a specialisation of machine learning that uses deep neural networks, characterised by a large number of hidden layers between the input and the output. The depth of these architectures is not merely a quantitative matter, but makes it possible to learn increasingly complex and abstract hierarchical representations. The initial layers tend to capture elementary and local features (such as edges and textures in images, or n-grams and syntactic patterns in texts), while the deeper layers combine these features in order to recognise high-level patterns, abstract concepts and complex semantic relations [6].

2.2.3 The training and validation process

The training of a machine learning model follows a rigorous and well-established methodology that involves the strategic division of the available data into three distinct sets, each with a specific role in the learning and evaluation process.

The **training set** is used for the actual learning, where the algorithm iteratively updates its internal parameters through an optimisation process that aims at minimising a loss function. This function quantifies the discrepancy between the predictions of the model and the desired target values, providing an error signal that guides the update of the weights through algorithms such as backpropagation and gradient descent.

The **validation set** plays a crucial role in monitoring the performance of the model on data not used for direct training. This set is employed in order to periodically evaluate the model during training, making it possible to identify the optimal moment at which to stop the training (early stopping) and to tune the hyperparameters of the model. The validation process is fundamental in order to avoid overfitting, a phenomenon in which the model excessively memorises the specific patterns of the training data, losing the ability to generalise to new examples.

The **test set** remains completely hidden throughout the whole development process and is used exclusively for the final evaluation of the performance. This set provides an unbiased and realistic estimate of the accuracy of the model on completely new data, simulating the real operating conditions in which the model will be used [7].

2.2.4 Transfer learning and fine-tuning

A fundamental concept for understanding the effectiveness and the computational economy of models such as BERT is **transfer learning**. This methodology makes it possible to transfer the knowledge acquired by a model during a general and computationally expensive task (pre-training) to specific applicative tasks (downstream tasks) through a specialisation process called fine-tuning [8].

In the specific case of BERT, the model initially undergoes a pre-training process on huge and diversified textual corpora through self-supervised tasks. This phase, which is extremely demanding in terms of computational and temporal resources (often requiring weeks of computation on specialised hardware such as GPUs or TPUs), allows the model to develop a deep and general understanding of linguistic structures, semantic relations and syntactic patterns.

Subsequently, through fine-tuning, this pre-trained model is specialised for specific tasks such as text classification, question answering or sentiment analysis. This process requires only a fraction of the original computational resources, typically a few hours or days, and makes it possible to adapt the general representations learned during pre-training to the specific needs of the target task.

Transfer learning is particularly powerful because it exploits the principle that many of the linguistic representations learned during pre-training (such as syntactic understanding, semantic relations, and discourse patterns) are intrinsically useful for a wide range of specific tasks. As will be shown in the experiments of this dissertation, a fine-tuned BERT model not only quantitatively improves the accuracy of the classification with respect to the base model, but also qualitatively restructures the embedding space, making the different classes semantically more separable and distinguishable.

2.2.5 Regularisation and generalisation

In order to guarantee that models develop a true generalisation ability rather than merely memorising the training data mechanically, machine learning employs several regularisation strategies. These techniques are particularly crucial in the context of large models, where the huge number of parameters can easily lead to overfitting phenomena.

Dropout represents one of the most effective and widely used regularisation techniques. During training, this methodology randomly deactivates a percentage of the neurons in every forward pass, forcing the model not to depend excessively on specific computational units and promoting the development of more robust and distributed representations [9].

Weight decay (or L2 regularisation) adds a penalty term to the loss function that discourages weights of large magnitude, favouring simpler and more generalisable solutions according to the principle of Occam’s razor. **Early stopping** monitors the performance on the validation set and interrupts the training when it stops improving, preventing overfitting before it manifests itself.

In the context of large language models such as BERT, which can contain hundreds of millions or billions of parameters, regularisation takes on critical importance. These models possess a representational capacity that is theoretically sufficient to completely memorise even very large training datasets, which makes regularisation techniques essential in order to guarantee that learning focuses on generalisable patterns rather than on idiosyncrasies specific to the training data.

2.3 The Transformer architecture

The revolution of LLMs was made possible by the introduction of the Transformer architecture, presented by Vaswani et al. in their celebrated work “*Attention Is All You Need*” [10]. Before the Transformer, natural language processing models were mainly based on recurrent (RNN) and long short-term memory (LSTM) architectures, which process text sequentially. This approach, although effective, presents two main limitations: the difficulty of parallelising the computations and the tendency to lose contextual information over long distances (the *vanishing gradient* problem).

The Transformer overcomes these obstacles by introducing the **self-attention** mechanism. Instead of processing words one after the other, the attention mechanism allows the model to weigh the importance of each word in a sequence with respect to all the others, regardless of their position. In this way, the model can create rich contextual representations, capturing long-range dependencies far more effectively.

The architecture is typically composed of two main blocks:

- **Encoder:** it processes the input sequence and transforms it into a continuous vector representation (embedding) that captures its contextual meaning.
- **Decoder:** it uses the representation produced by the encoder and the sequence generated so far in order to produce the next word in the output.

Models such as BERT, which focus on language understanding tasks, mainly use the encoder component.

2.4 The BERT model: pre-training and fine-tuning

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. [11], represents a milestone in the evolution of models based on the Transformer architecture. Its main innovation lies in introducing **bidirectional** training, which allows the model to consider simultaneously the right and left context of a word during the pre-training phase. This result is achieved through two unsupervised pre-training tasks: **Masked Language Model (MLM)** and **Next Sentence Prediction (NSP)**.

Once the expensive pre-training process has been completed, the BERT model can be adapted to specific tasks through a process called **fine-tuning**. In this phase, a small

classification layer is added on top of the BERT architecture and the whole model is trained for a limited number of epochs on a labelled dataset, specific to the target task. Fine-tuning makes it possible to transfer the deep linguistic knowledge acquired during pre-training to a specific problem. This process not only trains the new classification layer, but also updates the weights of the whole pre-trained Transformer model, specialising it in recognising the semantic nuances that are relevant for the task at hand. As shown in this work, a model fine-tuned for jailbreak/non-jailbreak classification is not only a classifier, but also becomes an embedding extractor that is semantically richer and more pertinent for that specific domain.

2.4.1 Extraction of semantic embeddings from BERT

The real power of models such as BERT lies in their ability to convert a text sequence into a fixed-size numerical vector, or **embedding**, that captures its contextual meaning. This vector is the internal representation that the model builds after processing the text.

In the context of BERT, a common practice consists in using the embedding associated with a special input token, the '[CLS]' token, which is always prepended to every sequence. Since the model is trained to use this token for sequence-level classification tasks (such as NSP), its final embedding (extracted from the last hidden layer of the Transformer) is considered an aggregated and semantically rich representation of the whole input sequence. It is precisely this technique that is employed in the experiments of this dissertation in order to obtain a single representative vector for each answer of the LLM.

2.5 Model alignment and jailbreaking

As the capabilities of LLMs increased, the need arose to guarantee a safe, ethical and useful behaviour. This process, known as **alignment**, has the purpose of teaching the model to follow a set of principles and guidelines. However, these safety barriers are not infallible. **Jailbreaking** is the act of deliberately circumventing the alignment measures of an LLM through deceptive queries, known as *jailbreak prompts*. This phenomenon represents a serious threat to the safety and the reliability of LLMs, making the development of methods to detect and mitigate it of fundamental importance.

2.6 Analysis through clustering with K-means

While supervised classification requires labelled data, unsupervised learning explores the data without pre-existing labels, seeking to discover latent structures. The **K-means algorithm** [12] is one of the best-known clustering techniques. Its objective is to partition a set of n observations (in this case, the embeddings of the answers) into k clusters, in which every observation belongs to the cluster with the nearest mean (centroid).

2.6.1 Evaluation of clustering quality

The choice of the optimal number of clusters, k , is a critical step. Over the course of the work carried out in this dissertation, two standard techniques were employed in order to guide this choice:

- **Elbow method:** it is based on the computation of the sum of the squared distances of every point from the centroid of its cluster (inertia). K-means is run for a range of values of k and the inertia is plotted as a function of k . The “elbow” in the plot, that is, the point at which the inertia stops decreasing rapidly, is considered an indicator of a good value for k .
- **Silhouette Score:** this metric measures how similar an object is to its own cluster with respect to the other clusters. The score ranges from -1 to 1, where a high value indicates that the object is well matched to its own cluster and poorly matched to the neighbouring clusters. Unlike inertia, a higher silhouette score is universally better, which makes it a more objective criterion for the selection of k , as done in the experimental code of this work.

The hierarchical approach adopted in the experiments, which first divides the data into two macro-clusters and then analyses each macro-cluster separately, makes it possible to refine the analysis and to discover finer substructures within the main “jailbreak” and “non-jailbreak” categories.

2.7 Dimensionality reduction techniques for visualisation

The embeddings generated by BERT have a high dimensionality (768 dimensions). In order to visualise and visually interpret the structure of the clusters, it is necessary to project them into a 2- or 3-dimensional space. In this work three standard techniques were used for this purpose:

- **PCA (Principal Component Analysis):** it is a linear technique that transforms the data into a new coordinate system in which most of the variance of the data lies along the first few coordinates (principal components). Fast and deterministic, PCA is excellent for obtaining a global view of the structure of the data, but it may fail to capture complex non-linear relations.
- **t-SNE (t-Distributed Stochastic Neighbor Embedding):** it is a non-linear technique that is particularly effective at visualising the local structure of the data, grouping similar points into well-separated clusters. It is highly sensitive to its hyperparameters (such as *perplexity*) and its computation is more demanding. The global distances between the clusters in a t-SNE plot are not necessarily meaningful.

- **UMAP (Uniform Manifold Approximation and Projection)**: it is another non-linear dimensionality reduction technique that is often considered a good compromise between PCA and t-SNE. It is generally faster than t-SNE and tends to preserve both the global and the local structure of the data better, which makes it a very popular choice for the visualisation of embeddings.

The combined use of these three techniques, as done in the experiments, provides a more robust and complete view of the separability and of the structure of the clusters in the embedding space.

CHAPTER 3

State of the art

The rapid proliferation of Large Language Models (LLMs) has brought with it a growing concern for their robustness and safety. The phenomenon of jailbreaking, in particular, has emerged as one of the main research areas. This chapter analyses the state of the art concerning jailbreaking attacks, defence techniques and, with particular attention, the methodologies for the **automatic evaluation** of the answers, in order to contextualise the contribution of this thesis.

3.1 Taxonomy of jailbreaking attacks

Jailbreaking attacks can be classified according to several parameters. The main categories, already explored in the literature, include:

- **Manual and template-based prompt engineering:** approaches based on human intuition, such as role playing (e.g. DAN) or the use of hypothetical scenarios.
- **Adversarial attacks and prompt optimisation:** algorithmic methods that automatically optimise the prompts in order to maximise the probability of success, both in white-box contexts (gradient-based) and in black-box ones (e.g. genetic algorithms [13] or iterative refinement such as PAIR [14]).
- **Encoding- and obfuscation-based attacks:** techniques that hide the harmful intent through unusual encodings, such as the use of low-resource languages [15] or obfuscation through symbolic mathematical problems (MathPrompt [16]).
- **Multimodal attacks:** they exploit the vulnerabilities of models that process both text and images, for example through “shuffle inconsistency” [17].

3.2 Defence mechanisms and evaluation platforms

In parallel with the attacks, research focuses on countermeasures and evaluation frameworks. Defences include input/output filtering, adversarial training and mechanisms

internal to the model such as the Activation Boundary Defense (ABD) [18]. In order to standardise the evaluation, benchmarks such as **JailbreakBench** [19] have been created, which provide common datasets and metrics for comparing attacks and defences.

3.3 Automatic evaluation of LLM answers

A crucial element, but one that is often taken for granted, in this attack-defence cycle is the ability to determine whether a given answer constitutes a jailbreak. Manual evaluation by human beings is considered the “gold standard” but it is slow, expensive, subjective and not scalable. This has pushed research towards automatic evaluation methods, which can be grouped into three main approaches.

3.3.1 Evaluation based on traditional metrics

Classical NLP metrics such as BLEU or ROUGE are useful for evaluating fluency or similarity to a reference text, but they are completely inadequate for evaluating the safety or the compliance of an answer. An answer can be grammatically perfect and fluent, but contain dangerous instructions. Therefore, these metrics are not applicable to the jailbreaking problem.

3.3.2 Evaluation through LLMs (LLM-as-a-Judge)

An emerging and powerful approach consists in using an advanced LLM (often a proprietary model such as GPT-4) as a “judge” in order to evaluate the answers of another model [20]. The judge model is given a prompt that describes the evaluation criteria (e.g. “Is the following answer harmful or evasive?”) and the answer to be judged.

- **Advantages:** this method is highly scalable and can capture complex nuances of language that simpler classifiers might miss.
- **Disadvantages:** it presents significant limitations. The judge model may have its own biases, it can be expensive to query, and above all it can be vulnerable to the same types of attack that it is supposed to detect. Moreover, the results may lack reproducibility because of the stochastic nature of the models and of the undocumented updates of the APIs.

3.3.3 Evaluation through specific classifiers

A more controlled and robust alternative consists in training a smaller and specialised classifier model. This is the approach that has been adopted in this thesis.

- **Methodology:** a dataset of answers manually labelled as “jailbreak” or “non-jailbreak” is created. On this dataset, a pre-trained model such as BERT is fine-tuned for the binary classification task.
- **Advantages:** these classifiers are fast, cheap to run, and their results are perfectly reproducible. Since they are specialised on a specific task, they can achieve very high performance. Unlike an LLM judge, a fine-tuned BERT classifier is a deterministic system whose functioning is transparent.

This methodology has become a de facto standard in many research pipelines on the security of LLMs, where fast and reliable feedback is required.

3.4 Limitations of the existing approaches and positioning of the thesis

From the analysis of the state of the art, some areas emerge that require further investigation.

1. **The quality of the embeddings:** although the use of BERT classifiers is common, the literature mainly focuses on their classification accuracy. There is less investigation of the quality of the semantic space (embedding space) that these models learn. Does a fine-tuned model create a space in which the classes are more separable with respect to a generic model?
2. **Beyond binary classification:** the evaluation often stops at a binary label (success/failure). There is a lack of large-scale quantitative analyses of the different *types* of jailbreak answers. Are there different “degrees” or “styles” of jailbreak that can be discovered automatically?

It is in this context that the present thesis work is placed. This research positions itself as a **methodological and comparative study on the automatic evaluation and on the unsupervised analysis of LLM answers**.

The contribution of this thesis is articulated on three levels, directly reflected in the experiments that were carried out:

- **Validation of the specific-classifier approach:** by using a fine-tuned BERT model (“Teto03/Bert_base_fineTuned”), the thesis adopts and validates the specific-classifier approach as the primary analysis tool.
- **Comparative analysis of the semantic spaces:** the innovative core of the thesis lies in the direct comparison between the embedding space generated by the **fine-tuned** BERT model and the one generated by a **base** BERT model (“bert-base-uncased”). The objective is to show quantitatively and qualitatively (through

clustering and visualisation) that fine-tuning not only improves the accuracy of the classification, but also restructures the semantic space, making “jailbreak” and “non-jailbreak” answers intrinsically more separable.

- **Unsupervised exploration of the answers:** going beyond classification, the application of a hierarchical clustering (K-means + silhouette analysis) aims at filling the second gap that was identified. This approach not only validates the separability of the macro-clusters (jailbreak vs. non-jailbreak), but also seeks to automatically discover thematic or stylistic subgroups within each category, providing deeper insight into the answering strategies of the target model.

In summary, this thesis does not limit itself to using an evaluation tool, but analyses its foundations, showing why a specialised model is superior to a generic one for semantic analysis in a specific domain, and proposing a combined methodology (classification + clustering) for a more complete and deeper evaluation of the security of LLMs.

CHAPTER 4

Methodology

4.1 Introduction

The present chapter describes in detail the methodology adopted in order to address the problem of the automatic classification of the answers generated by Large Language Models (LLMs), with particular attention to the identification of jailbreak attempts. As anticipated in the previous chapters, the increase in the architectural complexity and in the sophistication of LLMs has made it indispensable to adopt advanced techniques in order to detect behaviours that could turn out to be harmful or non-compliant with the ethical and functional alignment policies of the models.

The primary objective of the research is to develop an automatic evaluation system able to reliably distinguish between safe answers, which respect the ethical guidelines, and answers that could represent attempts to circumvent the restrictions imposed on the model, a phenomenon known as jailbreak. The main methodological challenge lies in the often subtle and nuanced nature of these circumvention attempts, which require approaches able to capture not only evident semantic differences, but also more subtle variations in the content, in the tone and in the intent of the answers.

The work is placed within the broader context of AI safety and of the alignment of language models, where the ability to automatically identify problematic behaviours represents a crucial element in guaranteeing a safe and ethical deployment of these systems. Methodological transparency and the validity of the procedures adopted are fundamental in order to guarantee the reproducibility and the reliability of the results.

4.2 Overview of the methodological approach

4.2.1 Evolution of the research framework

The methodology is articulated into a structured pipeline that underwent a significant evolution during the course of the research, moving from an initially rigid approach to a more sophisticated and flexible solution. This evolution was driven by the critical analysis

of the intermediate results and by the continuous feedback with the supervisors of the research, in particular with Prof. Ferretti and Prof. Saletta.

The methodological evolution can be divided into four main phases, each characterised by specific innovations and improvements that progressively refined the approach to the problem.

First phase - simplified binary classification approach (version 1.0)

The initial methodology was based on a direct binary classification approach, in which the fine-tuned BERT model produced an answer that was encoded as a two-dimensional vector. In this simplified representation, the X coordinate took discrete values in $\{-1, +1\}$, where $+1$ indicated “jailbreak” and -1 indicated “no-jailbreak”, while the Y coordinate represented the confidence of the model in the classification, with continuous values in the interval $[0, 1]$.

The operational flow of this version involved the processing of the answers by BERT, the generation of the classification with its associated confidence, the encoding into two-dimensional vectors and, finally, the application of the K-Means clustering algorithm with $k=2$ in order to group the results into two distinct categories.

Algorithm 1 Encoding of the classification result

Require: $classification \in \{0, 1\}$, $confidence \in [0, 1]$

Ensure: $data_point \in \{-1, 1\} \times [0, 1]$

```

1: // Encoding of the x coordinate based on the classification
2: if  $classification = 1$  then
3:    $x\_coord \leftarrow 1$  {Jailbreak}
4: else
5:    $x\_coord \leftarrow -1$  {Non-jailbreak}
6: end if
7: // Assignment of the y coordinate
8:  $y\_coord \leftarrow confidence$  {Confidence value between 0 and 1}
9: // Creation of the resulting two-dimensional vector
10:  $data\_point \leftarrow [x\_coord, y\_coord]$ 
11: return  $data\_point$ 

```

Although this initial setting offered immediate interpretability, the analysis carried out in collaboration with Prof. Ferretti highlighted several significant limitations. The two-dimensional representation turned out to be excessively rigid and reductive, unable to fully exploit the rich semantic information produced by BERT. The loss of information deriving from the reduction of the complexity of the problem to a rigid binary classification drastically limited the ability of the system to capture the nuances present in jailbreak attempts. Moreover, the clustering turned out to be suboptimal because of the low dimensionality of the generated vectors, which made it difficult to distinguish between borderline cases or sophisticated variations in the circumvention patterns.

It is important to stress that this first version, having an exploratory character, was implemented exclusively by using the fine-tuned BERT model and was tested on a significantly reduced dataset of only 20 examples. Such a limitation in the size of the dataset, although functional for the initial validation of the concept, contrasted sharply with the subsequent implementations, which are based on a substantially larger corpus of 1784 instances, distributed between 997 examples labelled as “no-jailbreak” (0) and 787 as “jailbreak” (1).

Development dataset The dataset used in the subsequent phases of the development comprises 1784 instances of prompts and their corresponding answers, carefully labelled for the binary classification of jailbreak attempts. The distribution of the labels turns out to be relatively balanced, with 997 instances classified as “no-jailbreak” (55.9%) and 787 instances classified as “jailbreak” (44.1%), guaranteeing a balanced representation of both categories for the training and the evaluation of the models in the subsequent versions of the architecture.

Second phase - extraction of high-dimensional embeddings (version 2.0)

In response to the limitations of the initial approach, an intermediate version was developed that fully exploited the richness of the BERT representations through the extraction of high-dimensional embeddings. This transition represents a fundamental paradigm shift in the methodological approach, motivated by the need to preserve the semantic richness of the representations generated by BERT.

The approach is based on the principles of the distributed representations of natural language, conceptually similar to Word2Vec proposed by Mikolov et al. (2013)[21], but applied at the level of the complete sequence. Unlike Word2Vec, which produces static representations for single words based on their co-occurrence in a corpus, BERT embeddings are contextual, with each representation varying as a function of the complete context of the sequence. These embeddings derive from the combination of 12 attention layers that capture different levels of linguistic abstraction and, in this specific case, they are optimised for jailbreak detection through fine-tuning.

The embedding extraction process follows a standardised procedure that can be formalised mathematically. Let text be a textual answer to be processed. The first step consists in the tokenisation by means of the **WordPiece** algorithm, which produces a sequence of tokens:

```
tokens = tokenizer(text, return_tensors="pt",
                  truncation=True, padding=True)
```

where tokens includes input_ids (numerical identifiers of the tokens), attention_mask (binary mask indicating the meaningful tokens) and optionally token_type_ids. Subsequently, the BERT model processes these inputs through its 12 transformer layers:

`outputs = model(**tokens, output_hidden_states=True)`

The hidden states H^l for each layer l , with $l = 0, 1, \dots, 12$, have size:

$$\text{batch_size} \times \text{seq_len} \times d$$

where $d = 768$ is the hidden size of BERT-base. The [CLS] vector of the last layer, used as the embedding of the sentence, is extracted as:

$$\mathbf{e} = H^{(12)}[:, 0, :] \in \mathbb{R}^{768}$$

This vector represents a synthesis of the contextual information of the whole sequence, weighted by the attention mechanism. The [CLS] token was chosen as the representation of the whole sequence because it is designed to aggregate global semantic information through the multi-head self-attention mechanism, where for a single head the attention is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q , K and V are the query, key and value matrices derived from the hidden states. The size of each head is given by:

$$d_k = \frac{d_{\text{model}}}{h} = \frac{768}{12} = 64$$

The use of 768-dimensional vectors offers several technical and linguistic advantages. Semantic richness is preserved by maintaining the linguistic and semantic nuances captured by the model. The embeddings also reflect the specific adaptation to jailbreak detection thanks to the sensitivity to fine-tuning, while the high-dimensional space guarantees greater freedom for the K-Means algorithm in the formation of coherent clusters.

Third phase - automatic clustering with optimisation of the number of clusters

The main limitation of the second version lay in fixing the number of clusters to $k=2$, which failed to capture the nuances and the gradations present in jailbreak datasets. The qualitative analysis of the answers, carried out in collaboration with Prof. Saletta, indeed showed the existence of intermediate categories and specific subtypes that required a more flexible approach.

Two main metrics were implemented for the automatic determination of the optimal number of clusters. The Elbow method is based on the analysis of the Within-Cluster Sum of Squares (WCSS), computed as:

$$\text{WCSS}(k) = \sum_{j=1}^k \sum_{x \in C_j} \|x - \mu_j\|^2$$

where C_j represents cluster j and μ_j its centroid. The optimal value of k is identified by the point at which the WCSS curve presents an “elbow”, indicating that further increases of k do not lead to significant improvements in the compactness of the clusters.

In parallel, a second method was also implemented, the Silhouette Analysis, which evaluates the quality of the clustering by computing the silhouette coefficient for every data point. For a point e_i , the coefficient $s(e_i)$ is defined as:

$$s(e_i) = \frac{b(e_i) - a(e_i)}{\max\{a(e_i), b(e_i)\}}$$

where $a(e_i)$ is the average distance of e_i from the other points of its cluster and $b(e_i)$ is the average distance of e_i from the nearest cluster to which it does not belong. The average silhouette score for a given k provides a global measure of the quality of the clustering, with higher values indicating a better definition of the clusters.

Fourth phase - two-level hierarchical clustering (final version)

The final methodology implements a two-level hierarchical clustering approach that combines the advantages of the previous phases, preserving the interpretability of the binary classification while capturing the semantic nuances through a multi-level analysis.

The hierarchical approach is articulated into two distinct levels. At the first level, K-Means with $k=2$ is applied in order to separate the main “jailbreak” versus “no-jailbreak” categories, maintaining the binary distinction that is useful for easily distinguishing the two macro-categories. At the second level, an automatic clustering is carried out within each macro-cluster in order to identify subtypes and variants, using the automatic optimisation of the number of clusters described in the previous phase.

This hierarchical approach presents numerous methodological advantages. It maintains interpretability by preserving the binary distinction that is fundamental for practical security purposes, while it increases granularity by identifying semantically coherent subcategories within each macro-category. The adaptive flexibility allows the automatic determination of the optimal number of subclusters for each macro-cluster, reflecting the possible asymmetry in the internal complexity of the two main categories. Moreover, the multi-scale validation allows the application of quality metrics at both levels of the hierarchy, guaranteeing statistical robustness and semantic coherence.

4.3 Experimental setup and models used

4.3.1 Dual-model comparative architecture

In order to guarantee the robustness and the validity of the results, the methodology involves the use of two parallel pipelines based on distinct BERT models, allowing a comparative analysis of the effectiveness of the semantic representations and of the impact of specialised fine-tuning.

The first model used is BERT-base-uncased, which represents the generalist baseline. It is the standard model pre-trained on the traditional BookCorpus and English Wikipedia corpora, with 110 million parameters distributed over 12 transformer layers and 768 hidden state dimensions. Its role is to provide general-purpose semantic representations without specific biases for the jailbreak domain, making it possible to evaluate the effectiveness of specialised approaches with respect to a generalist model.

The second model is Teto03/Bert_base_fineTuned, developed by my colleague Brighenti Stefano specifically for jailbreak detection. This model was fine-tuned on a curated dataset containing representative examples of circumvention techniques, resulting in representations optimised to distinguish the linguistic patterns associated with jailbreak attempts. The specialisation offers greater sensitivity to the specific characteristics of the domain of interest, such as the use of indirect language, metaphors to circumvent restrictions, or formulation patterns that typically precede non-compliant answers.

The choice of using both models is motivated by the need to empirically quantify the contribution of specialised fine-tuning with respect to the generalist capabilities of standard pre-trained models. This comparative configuration also makes it possible to evaluate the generalisability of the results and to identify any biases introduced by the fine-tuning process.

4.3.2 Evaluation dataset and preprocessing

The dataset used for the evaluation consists of a curated selection derived from the files of LLM answers provided by Prof. Ferretti. It is crucial to stress that these data were not used during the fine-tuning phase of the model, thus guaranteeing that the data are completely new to the model and avoiding overfitting phenomena. This rigorous separation allows an authentic evaluation of the generalisation capabilities of the model on data not seen during training, preventing predetermined answers to questions already processed by the model.

The preprocessing of the data was carried out in close collaboration with Brighenti Stefano and followed a rigorous pipeline in order to ensure the quality and the uniformity of the answers. The process includes several crucial phases, starting with the cleaning of the data, which comprises the removal of special characters and inconsistent formatting, the normalisation of spacing and punctuation, and the appropriate handling of encoding

and Unicode characters. When necessary, a machine translation phase was applied in order to guarantee linguistic uniformity, with the subsequent standardisation of the format of the answers and validation of the quality of the translations. The final phase involves a manual check of a representative sample in order to verify the semantic coherence after preprocessing and to remove any corrupted or incomplete answers.

4.3.3 Experimental configuration parameters

The experimental configuration was standardised in order to guarantee the reproducibility and the comparability of the results. The fundamental parameters include a reproducibility seed fixed at 42, used consistently for all the stochastic algorithms in order to guarantee deterministic results. The maximum number of iterations for the K-Means algorithm was set to 300, guaranteeing convergence even for complex configurations. The range of clusters for the automatic optimisation was defined as $k \in [2, 8]$, balancing the ability to capture meaningful subtypes with the need to keep the clusters interpretable. The maximum sequence length was fixed at 512 tokens, corresponding to the standard limit of BERT, while the batch size for inference is managed dynamically according to the available memory, optimising computational efficiency.

As regards the dimensionality reduction parameters, PCA is configured with `n_components=2` and `random_state=42` in order to guarantee a reproducible linear projection. t-SNE uses `n_components=2` with the perplexity adapted dynamically as `min(30, n_samples-1)` in order to handle datasets of variable size. UMAP is set with `n_components=2` and `random_state=42`, using default values for `neighbors` and `min_dist`, which have shown good performance in the literature for similar tasks.

4.4 Extraction and representation of the embeddings

4.4.1 Detailed extraction process

The embedding extraction process represents the core of the methodology, transforming the textual answers into numerical representations suitable for clustering. This process begins with a systematic preprocessing phase that prepares the texts for processing by the transformer models.

The tokenisation uses the WordPiece algorithm for the decomposition into sub-lexical units, effectively handling out-of-vocabulary words and preserving the relevant morphological information. During this process, the special tokens [CLS] at the beginning and [SEP] at the end of every sequence are added automatically. The [CLS] token acts as an aggregator of global semantic information, accumulating through the layers of the transformer a compressed representation of the whole sequence. The sequences are made uniform through padding for those shorter than 512 tokens and intelligent truncation for those exceeding this limit, preserving the central semantic information.

The actual extraction takes place by processing the tokenised sequences through the BERT model. For every answer in the dataset, we obtain an embedding e_i computed as:

Algorithm 2 Extraction of CLS embeddings

Require: *text*: input string, *model*: transformer model, *tokenizer*: tokeniser

Ensure: *cls_embedding*: CLS embedding as a numpy array

```

1: function EXTRACT_CLS_EMBEDDING(text, model, tokenizer)
2: // Tokenisation of the input text
3: inputs  $\leftarrow$  TOKENIZER(text, return_tensors="pt", truncation=True,
4: padding=True, max_length=512)
5: // Extraction of the model outputs without gradient computation
6: with NO_GRAD do
7:   outputs  $\leftarrow$  MODEL(inputs, output_hidden_states=True)
8:   // Extraction of the CLS embedding from the last hidden state
9:   cls_embedding  $\leftarrow$  outputs.hidden_states[-1][:, 0, :]
10: end with
11: // Conversion to numpy format and detachment from the gradients
12: cls_embedding  $\leftarrow$  cls_embedding.DETACH().CPU().NUMPY()
13: return cls_embedding
14: end function

```

By repeating this process for all the N answers of the dataset, we obtain an embedding matrix $X \in \mathbb{R}^{N \times 768}$ where every row represents the embedding of one answer.

4.4.2 Quality and properties of the embeddings

BERT embeddings present superior characteristics with respect to traditional static representations. Dynamic contextuality guarantees that every embedding varies as a function of the complete context, capturing semantic nuances that would otherwise be impossible to detect. For example, ambiguous expressions that could indicate jailbreak attempts are represented differently according to the surrounding context, allowing a more accurate discrimination.

The multilayer abstraction combines information from 12 levels of transformation, where the first layers mainly capture syntactic and morphological features, while the upper layers encode increasingly abstract semantic and pragmatic relations. This hierarchy of representations is particularly useful for jailbreak detection, where superficial patterns can mask deeper intents.

The sensitivity to fine-tuning in the case of the Teto03/Bert_base_fineTuned model means that the embeddings were optimised specifically in order to emphasise the distinctive characteristics of jailbreak attempts. During fine-tuning, the model learned to give greater weight to linguistic patterns such as the use of indirect or metaphorical language to circumvent restrictions, formulations that attempt to reinterpret illegal requests in apparently harmless ways, or argumentative structures that seek to justify non-compliant answers.

This representational richness is fundamental for the accurate identification of jailbreaks, where circumvention attempts often exploit semantic ambiguities, require an understanding of the pragmatic context, and may present superficial stylistic variations that should not influence the substantial classification.

4.5 Clustering algorithms and optimisation

4.5.1 K-Means: foundations and implementation

The K-Means algorithm was selected as the main clustering method for its computational efficiency, with complexity $O(Nkt)$, where N is the number of points, k the number of clusters and t the number of iterations. Convergence is mathematically guaranteed, although to a local optimum, and the resulting centroids provide interpretable exemplar representations of the clusters. The algorithm has also shown robustness on high-dimensional spaces such as BERT embeddings.

The multi-level implementation follows a hierarchical strategy. At the first level, the main clustering with $k=2$ separates the answers into the fundamental categories through the minimisation of the objective function:

$$J = \sum_{i=1}^N \sum_{j=1}^2 r_{ij} \|x_i - \mu_j\|^2$$

where r_{ij} is the binary assignment of point i to cluster j . After convergence, a semantic mapping of the clusters is carried out based on the qualitative inspection of representative samples, assigning the label “jailbreak” to the cluster that predominantly contains problematic answers.

At the second level, for each macro-cluster a sub-clustering is applied with the number of clusters determined automatically. The optimisation is based on the maximisation of the average Silhouette Score, which provides a balanced measure of internal cohesion and separation between clusters.

4.5.2 Optimisation metrics

The Silhouette Score for a single point measures how well that point fits its own cluster with respect to the others. A value close to 1 indicates that the point is well assigned, while negative values suggest a possible erroneous classification. The geometric interpretation of the coefficient reveals that maximising the Silhouette Score is equivalent to finding compact and well-separated clusters in the embedding space.

The Elbow method provides a complementary perspective by analysing how the intra-cluster inertia decreases as the number of clusters increases. The “elbow” point represents

an optimal trade-off between the complexity of the model (number of clusters) and the quality of the representation (compactness of the clusters). Mathematically, we look for the point at which the second derivative of the WCSS curve changes sign most markedly.

4.5.3 Validation and quality metrics

Validation is applied systematically at both levels of the hierarchical clustering. At the macro level, the validation focuses on verifying that the binary jailbreak/no-jailbreak separation is semantically coherent and statistically significant. At the micro level, the attention is on the internal coherence of the subclusters and on their interpretability in terms of subtypes of jailbreak or of safe answers.

The integrated metrics that are used include the Silhouette Score for the overall quality, the **Calinski-Harabasz Index**, which measures the ratio between the between-cluster and the within-cluster variance, and the **Davies-Bouldin Index**, which quantifies the average similarity between clusters, where lower values indicate better separation. These metrics provide complementary perspectives on the quality of the clustering, allowing a robust evaluation of the results.

4.6 Visualisation and interpretation

4.6.1 Dimensionality reduction techniques

The visualisation of high-dimensional embeddings requires dimensionality reduction techniques that preserve the essential structural relations. PCA provides a linear projection that is optimal in the least-squares sense, maximising the variance preserved in the first components. Mathematically, PCA finds the directions v that maximise:

$$\text{var}(Xv) \quad \text{subject to} \quad \|v\| = 1$$

The resulting principal components provide an interpretable decomposition of the variance in the data, making it possible to identify the directions of maximum variation in the embeddings.

t-SNE implements a non-linear reduction that preserves local neighbourhood relations by minimising the **Kullback-Leibler** divergence between the probability distributions in the original and in the projected space. The cost function of t-SNE is:

$$C = \sum_i \sum_j p_{ij} \log(p_{ij}/q_{ij})$$

where p_{ij} represents the similarity between points in the original space and q_{ij} in the projected space. This optimisation produces visualisations that emphasise the local structure of the clusters.

UMAP extends these concepts by using the theory of algebraic topology in order to preserve both the local and the global structure of the data. The algorithm builds a fuzzy graph in the high-dimensional space and optimises a low-dimensional representation that preserves the topology of this graph.

4.6.2 Multi-level visualisations

The visualisations were designed so as to reflect the hierarchical structure of the clustering. The visualisation of the macro-clusters highlights the fundamental binary separation, with the projected centroids providing visual references for the prototypes of each category. The complete hierarchical visualisation integrates both levels, using a colour coding that reflects both the membership in the macro-cluster and in the sub-cluster, making it possible to appreciate simultaneously the global structure and the local distinctions.

4.7 Validation and robustness

4.7.1 Validation protocol

The validation protocol combines quantitative and qualitative approaches in order to guarantee the robustness of the results. The internal validation focuses on the statistical stability of the clustering, verified through multiple runs with different initialisations of the centroids. The convergence analysis monitors the number of iterations required and the stability of the final assignments. The sensitivity to the parameters is evaluated systematically by varying the key parameters such as the maximum number of iterations and the convergence criteria.

The external validation compares the results obtained with BERT-base and with fine-tuned BERT, quantifying the impact of specialised fine-tuning through metrics such as the Adjusted Rand Index and the Normalized Mutual Information. The qualitative inspection of representative samples for every cluster verifies the semantic coherence of the assignments and identifies any patterns not captured by the quantitative metrics.

4.7.2 Handling of uncertainty

The analysis of the statistical stability quantifies the variability of the results through repeated runs, providing confidence intervals for the quality metrics. The sensitivity to the dimensionality reduction techniques is evaluated by systematically comparing the results obtained with PCA, t-SNE and UMAP, verifying that the identified structures are not artefacts of a specific projection.

4.8 Final implementation pipeline

4.8.1 System architecture

The final pipeline integrates all the methodological components into a coherent and modular flow. The architecture was designed so as to maximise reproducibility while maintaining the flexibility needed for future extensions. The main function `JailbreakClusteringPipeline` encapsulates the whole process, from the extraction of the embeddings to the visualisation of the results.

The implementation of two parallel pipelines, one for BERT-base and one for fine-tuned BERT, allows a systematic comparative analysis. This dual-model architecture represents a fundamental methodological element in order to quantify the improvement in the jailbreak detection capabilities obtained through training on domain-specific data.

The operational flow follows a structured sequence that guarantees reproducibility and modularity. The LLM answers are processed through the tokeniser specific to the model, and converted into numerical representations compatible with the transformer architecture. The tokenisation process preserves the essential semantic characteristics while preparing the text for processing. Subsequently, the model processes the sequences, producing high-dimensional vector representations through the multi-layer self-attention mechanism.

4.8.2 Data management and preprocessing

The dataset that was used derives from a curated selection of the LLM answers provided by Prof. Ferretti. The rigorous separation between training and testing data guarantees an unbiased evaluation of the generalisation capabilities of the system. The preprocessing required particular attention given the heterogeneous nature of the LLM answers, and included phases of normalisation, handling of special characters and, when necessary, machine translation for linguistic uniformity.

The architecture of the pipeline was designed with particular attention to scalability. The extraction of the embeddings takes place in batch mode whenever possible, with dynamic management of the size based on the available memory. The caching of the extracted embeddings represents an important optimisation, making it possible to reuse the representations for different clustering configurations without having to run the expensive extraction phase again.

4.9 Methodological conclusions

The methodology that was developed represents an innovative approach to the problem of the automatic classification of jailbreaks in LLM answers. The evolution from the initial rigid binary classification to the final hierarchical system reflects an iterative process of

refinement guided by the empirical analysis of the results and by the feedback of the domain experts.

The integration of deep learning techniques for feature extraction with unsupervised clustering algorithms makes it possible to combine the ability of transformer models to capture complex semantics with the flexibility of unsupervised approaches in identifying emergent structures in the data. The two-level hierarchical clustering effectively balances the practical needs of binary classification with the need to capture the nuances present in real jailbreak attempts.

The multi-perspective validation through different visualisation techniques and quality metrics guarantees the robustness of the results and facilitates the interpretation by human operators. The parallel implementation with base and fine-tuned models also provides valuable insight into the importance of specialised training for this specific application domain. The results obtained, which will be presented in the next chapter, demonstrate the effectiveness of the proposed approach in detecting and categorising jailbreak attempts with high accuracy and granularity.

CHAPTER 5

Presentation and analysis of the results

5.1 Introduction

This chapter presents a critical and in-depth analysis of the results obtained through the methodological evolution of the clustering system for the classification of jailbreak answers. The approach that was developed went through several refinement phases, from the initial simplified version to the final hierarchical system, making it possible to empirically evaluate the effectiveness of the different strategies that were adopted. The analysis focuses both on the quantitative improvements in the performance and on the qualitative interpretability of the results, providing an overall evaluation of the methodological contribution and of its implications for the security of language models.

The discussion proceeds through a systematic comparison between the different versions that were implemented, highlighting the strengths and the limitations of each approach. Particular attention is devoted to the analysis of the clustering quality metrics and to the validation through dimensional visualisation techniques.

5.2 Results of version 1.0: simplified two-dimensional approach

5.2.1 Methodological configuration and initial results

The initial version of the system represented a first attempt at applying clustering techniques directly to the BERT classification results. The approach used an extremely simplified two-dimensional representation, in which every answer of the evaluation dataset was encoded through two coordinates: the first represented the discrete binary classification (no-jailbreak = -1, jailbreak = +1), while the second captured the confidence level of the model as a continuous value in the interval $[0,1]$.

The analysis of the results obtained on an initial sample of 20 answers highlighted interesting but limited patterns. The distribution of the classifications showed a perfect balance, with 10 elements classified as no-jailbreak and 10 as jailbreak.

Table 5.1: Summary results of version 1.0 on a sample of 20 answers

Metric	No-Jailbreak (X = -1)	Jailbreak (X = +1)
Number of answers	10	10
Average confidence	0.943	0.977
Standard deviation	0.156	0.025
Confidence range	0.516 - 0.999	0.953 - 0.999
Assigned cluster	Cluster 0	Cluster 1

The analysis of the confidence revealed significant differences between the two categories: the answers classified as jailbreak showed an average confidence of 0.977 with a standard deviation of 0.025, while the no-jailbreak ones presented an average confidence of 0.943 with a higher standard deviation of 0.156. The confidence range for the jailbreak answers extended from 0.953 to 0.999, indicating a greater certainty of the model in the identification of these patterns. On the contrary, the no-jailbreak answers showed a wider range (0.516-0.999), suggesting a greater variability in the classification certainty for this category.

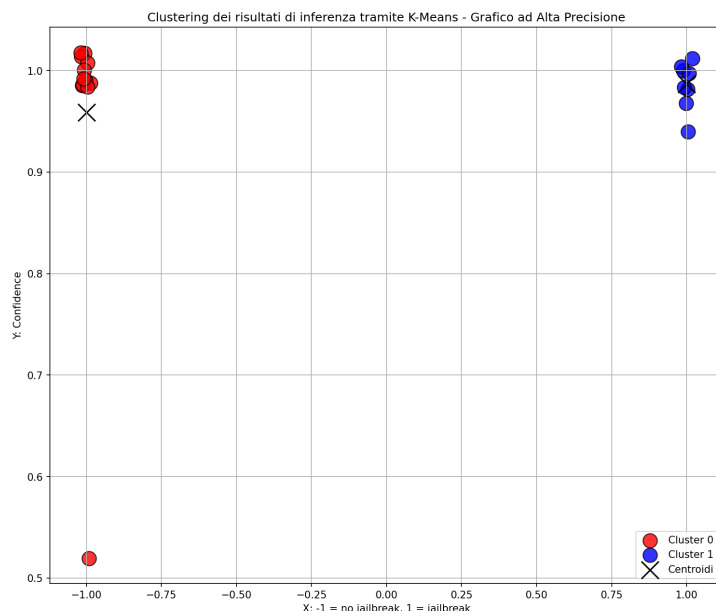


Figure 5.1: Two-dimensional distribution of the results of version 1.0. The X axis represents the binary classification (-1 = no-jailbreak, +1 = jailbreak), while the Y axis shows the confidence of the model.

5.2.2 Limitations of the two-dimensional approach

The application of the K-Means algorithm with $k=2$ on this representation produced predictable but not very informative results. The separation of the clusters took place perfectly along the X axis, essentially replicating the original BERT classification without adding interpretative value. This algorithmic redundancy represented one of the main limitations of the initial approach.

The loss of semantic information constituted the most critical problem of version 1.0. The reduction of the 768-dimensional embeddings to only two coordinates entailed a drastic loss of the semantic nuances captured by the BERT model. Moreover, the impossibility of identifying subtypes or gradations within the main categories significantly limited the analytical usefulness of the system.

5.3 Results of version 2.0: high-dimensional embeddings with fixed clustering

5.3.1 Transition to complete embeddings

The recognition of the limitations of version 1.0 guided the development of version 2.0, characterised by the direct use of 768-dimensional BERT embeddings with clustering fixed at $k=2$. This transition represented a fundamental paradigm shift, motivated by the need to preserve the semantic richness of the representations generated by the transformer model.

5.3.2 Dataset and original composition

The complete dataset used for this analysis is composed of **1,784** answers manually labelled into two categories:

- **997 examples labelled as 0 (no break) - safe answers compliant with the guidelines (55.9%)**
- **787 examples labelled as 1 (break) - jailbreak attempts or problematic answers (44.1%)**

This original distribution presents a slight imbalance towards the “no break” class, reflecting a realistic composition for operational AI security scenarios, where jailbreak attempts constitute a significant but not predominant fraction of the total traffic.

5.3.3 Clustering results on the complete dataset

The application of K-Means clustering with a fixed $k=2$ on the complete dataset of 1,784 answers produced significantly more informative results with respect to the previous

version. Table 5.2 shows a direct comparison between the two models.

Table 5.2: Comparison of the cluster distributions: BERT Fine-tuned vs BERT Base

Model	Cluster 0	Percentage	Cluster 1	Percentage
BERT Fine-tuned	1,087	61.31%	686	38.69%
BERT Base	331	18.67%	1,442	81.33%

Critical observations on the distribution:

- The fine-tuned model produces a balanced distribution (61.31% vs 38.69%) that better reflects the composition of the original dataset (55.9% vs 44.1%)
- The base model shows a strong asymmetry, with more than 80% of the points concentrated in Cluster 1
- This difference indicates that fine-tuning has restructured the embedding space towards a more natural separation of the categories

5.3.4 Analysis of accuracy and correspondence with the ground truth

In order to evaluate the effectiveness of the clustering, a systematic analysis of the correspondence between the identified clusters and the ground truth labels was carried out. Since clustering is unsupervised, it is necessary to determine the optimal mapping between mathematical clusters and semantic categories.

Table 5.3: Accuracy of the clustering with respect to the original labels

Model	Optimal correspondence	Accuracy
BERT Fine-tuned	Cluster 0 = No break, Cluster 1 = Break	84.7%
	Cluster 0 = Break, Cluster 1 = No break	46.8%
BERT Base	Cluster 0 = No break, Cluster 1 = Break	68.9%
	Cluster 0 = Break, Cluster 1 = No break	54.2%

Key results:

- Fine-tuning produces an improvement of 15.8 percentage points in the accuracy (from 68.9% to 84.7%)
- For both models, the optimal correspondence is Cluster 0 = No break, Cluster 1 = Break
- The substantial difference between the two accuracy percentages indicates a well-defined cluster structure in the fine-tuned model

5.3.5 Detailed analysis with confusion matrices

For a deeper understanding of the performance, we analyse the distribution of the original labels in the clusters generated by the two models.

Table 5.4: Confusion matrix - BERT Fine-tuned

Original label	Cluster 0 (No break)	Cluster 1 (Break)	Total
No break (0)	891 (89.4%)	106 (10.6%)	997 (100%)
Break (1)	196 (24.9%)	591 (75.1%)	787 (100%)
Total	1,087	686	1,784

Table 5.5: Confusion matrix - BERT Base

Original label	Cluster 0 (No break)	Cluster 1 (Break)	Total
No break (0)	307 (30.8%)	690 (69.2%)	997 (100%)
Break (1)	24 (3.0%)	763 (97.0%)	787 (100%)
Total	331	1,442	1,784

Interpretation of the confusion matrices:

BERT FINE-TUNED:

- High specificity for “no break”: 89.4% of the safe examples are correctly assigned to Cluster 0
- Good sensitivity for “break”: 75.1% of the jailbreak examples are identified in Cluster 1
- Optimal balance: the model shows balanced discriminative capabilities for both categories

BERT BASE:

- Excellent sensitivity for “break”: 97.0% of the jailbreak examples are identified
- Critical specificity for “no break”: only 30.8% of the safe examples are correctly classified
- Severe imbalance: the model tends to over-classify everything as a potential “break”

5.3.6 Clustering quality metrics

In order to quantify the quality of the separation objectively, standard clustering evaluation metrics were computed.

Table 5.6: Separability and compactness metrics of the clusters

Model	Centroid distance	Average Silhouette Score
BERT Fine-tuned	2.45	0.32
BERT Base	1.12	0.15

Interpretation of the metrics:

- Centroid distance: the fine-tuned model presents an inter-cluster separation that is 2.19 times higher
- These values quantitatively confirm the qualitative superiority observed in the visualisations

5.3.7 Validation through dimensional visualisations

For this version of the system, the validation was also carried out with the aid of dimensionality reduction techniques. Specifically, PCA (Principal Component Analysis) and t-SNE (t-Distributed Stochastic Neighbor Embedding) were employed.

Comparison of the PCA projections

The two-dimensional PCA projection highlighted substantial differences between the two models. The fine-tuned BERT model shows an extremely sharp linear separation, while the base model presents a significantly less defined separation.

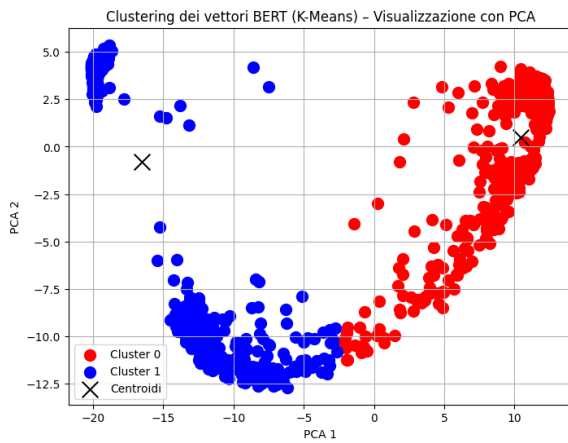


Figure 5.2: PCA projection with BERT Fine-tuned. Extremely sharp separation with well-defined geometries.

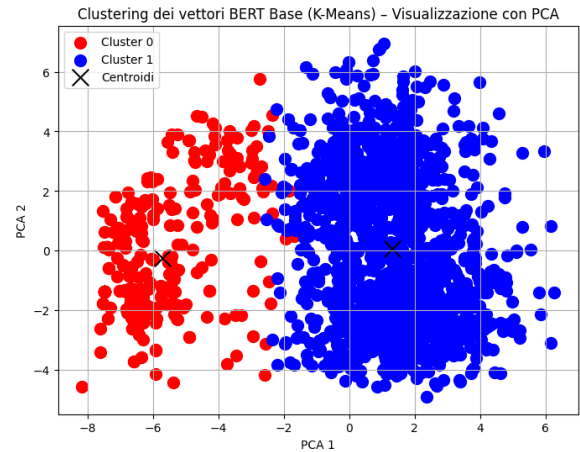


Figure 5.3: PCA projection with BERT Base. Less sharp separation with greater overlap.

Key observations from the PCA projections:

- Fine-tuned model: the clusters form two compact and linearly separable regions with sharp boundaries
- Base model: greater overlap and dispersion, indicating a less organised structure of the embedding space
- The first two principal components in the fine-tuned model explain a larger percentage of the total variance

Comparison of the t-SNE projections

The t-SNE analysis reveals even more marked differences between the two approaches. The fine-tuned model produces extremely compact and well-separated clusters, while the base model shows greater dispersion and fragmentation.

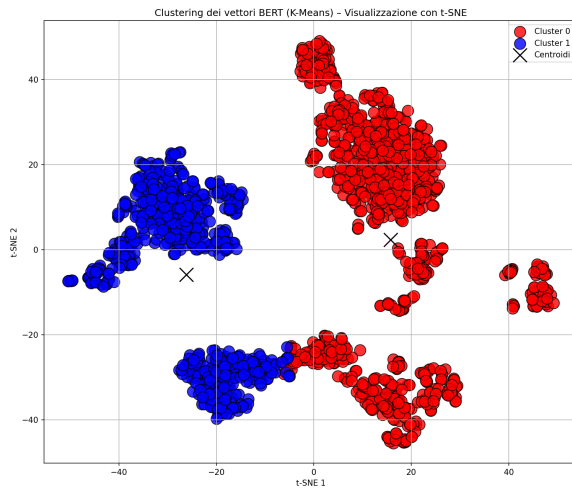


Figure 5.4: t-SNE projection with BERT Fine-tuned. Extremely compact and well-separated clusters.

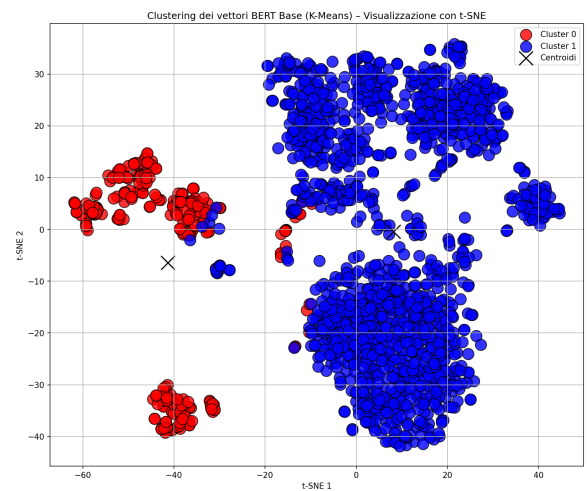


Figure 5.5: t-SNE projection with BERT Base. Greater dispersion and internal fragmentation.

Observations from the t-SNE visualisations:

- Intra-cluster compactness: the fine-tuned model generates much denser and more cohesive clusters
- Inter-cluster separation: the distance between the cluster regions is substantially greater in the specialised model
- Local structure: the local similarity relations are better preserved in the fine-tuned model

5.3.8 Interpretation of the differences

The visual and quantitative comparison between the two models provides empirical evidence of the transformative impact of specialised fine-tuning. The observed differences can be interpreted from several perspectives:

Restructuring of the semantic space: fine-tuning has evidently modified the geometry of the embedding space, making the jailbreak and non-jailbreak categories more linearly separable. This transformation is not merely quantitative but represents a qualitative reorganisation of the semantic representations that optimises discriminability for the specific task.

Greater intra-cluster cohesion: the clusters of the fine-tuned model show a significantly higher internal cohesion (Silhouette Score: 0.32 vs 0.15), indicating that the specialisation process has emphasised the semantic similarities within each category, creating more homogeneous representations for examples of the same class.

Improved inter-cluster separability: the distance between the centroids of the clusters (2.45 vs 1.12) and the sharpness of the boundaries in the visualisations suggest that fine-tuning has amplified the semantic differences between the target categories, making the automatic distinction more reliable and robust.

More realistic distribution: the more balanced distribution in the fine-tuned model (61.31% vs 38.69%) with respect to the base model (18.67% vs 81.33%) indicates a better calibration for the specific domain. The specialised model produces clusters that more faithfully reflect the natural composition of the ground truth dataset.

Balanced precision vs recall: while the base model shows a high sensitivity for jailbreaks (97.0%) but poor specificity for safe answers (30.8%), the fine-tuned model reaches an optimal balance (89.4% specificity, 75.1% sensitivity).

5.3.9 Implications for the effectiveness of the system

These results empirically confirm the hypothesis that specialised fine-tuning not only improves the classification accuracy, but also fundamentally transforms the quality of the semantic representations for the specific domain. The evidence shows that:

- The fine-tuned model generates more discriminative embeddings for the jailbreak/non-jailbreak distinction
- The geometric structure of the embedding space turns out to be optimised for the specific task

- The quality of the clustering improves substantially through the specialisation of the model
- The representations become more interpretable and reliable for security applications
- The balance between precision and recall is optimised for realistic operational scenarios

This evidence strongly supports the methodological approach that was adopted and justifies the computational investment required for specialised fine-tuning in the context of the security of language models, preparing the ground for the evolution towards the final hierarchical system that will be described in the following sections.

5.4 Results of version 3.0: adaptive clustering with automatic determination of the optimal number of clusters

5.4.1 Transition to adaptive clustering

Version 3.0 represents a significant methodological evolution with respect to the previous approach, eliminating the constraint of clustering fixed at $k=2$ and implementing techniques for the automatic optimisation of the number of clusters. This transition is motivated by the hypothesis that the intrinsic semantic structure of the dataset may require a segmentation granularity higher than the binary one, potentially revealing latent semantic subcategories within the main classes.

The process of automatic determination of the optimal number of clusters was carried out through the combined analysis of the Elbow method and of the Silhouette Score, making it possible to identify the configuration that simultaneously maximises the intra-cluster cohesion and the inter-cluster separation.

5.4.2 Dataset and optimisation methodology

The dataset that was used maintains the same composition as in the previous version:

- 997 examples labelled as 0 (no break) - safe answers compliant with the guidelines (55.9%)
- 787 examples labelled as 1 (break) - jailbreak attempts or problematic answers (44.1%)
- Total: 1,784 manually labelled answers

5.4.3 Results of the automatic clustering

The application of the optimisation algorithm produced substantially different configurations for the two models, highlighting the impact of fine-tuning on the geometric structure of the embedding space.

Table 5.7: Cluster distribution: automatic vs fixed clustering

Model	Optimal clusters	Cluster 0	Cluster 1	Cluster 2	Silhouette
BERT Fine-tuned	3	1,060 (59.79%)	453 (25.55%)	260 (14.66%)	0.61
BERT Base	2	331 (18.67%)	1,442 (81.33%)	-	0.20

Observations on the automatic distribution:

- The fine-tuned model automatically identifies 3 optimal clusters, revealing latent semantic substructures
- The Silhouette Score improves drastically from 0.20 in the base model to 0.61 in the fine-tuned model
- The tri-cluster distribution suggests the existence of semantically distinct jailbreak subcategories

5.4.4 Analysis of the optimisation metrics

In order to evaluate the effectiveness of the optimisation process, the Elbow and Silhouette curves were analysed for both models.

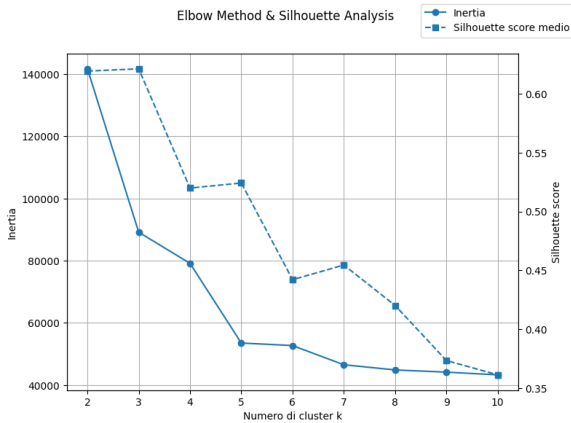


Figure 5.6: Elbow and Silhouette analysis - BERT Fine-tuned. The optimal point at $k=3$ is clearly identifiable, with a maximum Silhouette Score of 0.61.

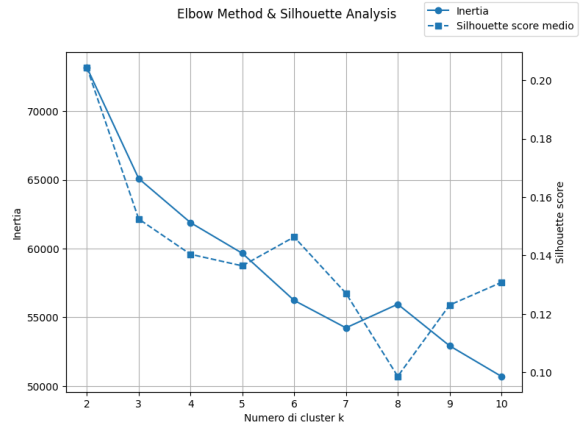


Figure 5.7: Elbow and Silhouette analysis - BERT Base. Convergence towards $k=2$ with a maximum Silhouette Score limited to 0.20.

Table 5.8: Detailed optimisation metrics for the range k=2-10

k	BERT Fine-tuned		BERT Base	
	Inertia	Silhouette	Inertia	Silhouette
2	142,000	0.62	73,000	0.20
3	90,000	0.61	65,000	0.16
4	80,000	0.52	62,000	0.14
5	54,000	0.53	59,000	0.14
6	52,000	0.44	56,000	0.15
7	47,000	0.46	54,000	0.13
8	45,000	0.42	51,000	0.10
9	44,000	0.37	53,000	0.12
10	43,000	0.36	51,000	0.13

Interpretation of the optimisation metrics:

- BERT Fine-tuned: optimum clearly identifiable at k=3 with a Silhouette Score of 0.61, closely followed by k=2 (0.62)
- BERT Base: global maximum at k=2 with a Silhouette Score limited to 0.20, indicating a natural binary structure but of poor quality
- Qualitative improvement: fine-tuning produces an improvement of 3.05x in the optimal Silhouette Score
- Stability of the configuration: the fine-tuned model maintains high Silhouette Scores for k=2-3, indicating structural robustness

5.4.5 Analysis of accuracy and correspondence with the ground truth

In order to evaluate the effectiveness of the automatic clustering, a systematic analysis of the correspondence between the identified clusters and the ground truth labels was carried out, considering the optimal mapping for both configurations.

Table 5.9: Accuracy of the automatic clustering with respect to the original labels

Model	Optimal configuration	Accuracy
BERT Fine-tuned (k=3)	Cluster 0=No break, Cluster 1+2=Break	84.5%
	Alternative mapping	32.1%
BERT Base (k=2)	Cluster 0=No break, Cluster 1=Break	59.7%
	Cluster 0=Break, Cluster 1=No break	40.3%

Key results on the accuracy:

- The adaptive clustering maintains a high accuracy (84.5%) while revealing additional structures
- Substantial improvement with respect to the base model: +24.8 percentage points
- The tri-cluster configuration does not compromise the binary performance but adds semantic granularity
- The base model presents a deterioration with respect to version 2.0 (-9.2 points), suggesting instability in the automatic configuration

5.4.6 Detailed analysis with confusion matrices

For a deep understanding of the semantic correspondence, we analyse the distribution of the original labels in the automatically identified clusters.

Table 5.10: Confusion matrix - BERT Fine-tuned (3 clusters)

Original label	Cluster 0	Cluster 1	Cluster 2	Total
No break (0)	891 (89.4%)	106 (10.6%)	0 (0.0%)	997 (100%)
Break (1)	169 (21.5%)	347 (44.1%)	271 (34.4%)	787 (100%)
Total	1,060	453	271	1,784

Table 5.11: Confusion matrix - BERT Base (2 clusters)

Original label	Cluster 0	Cluster 1	Total
No break (0)	307 (30.8%)	690 (69.2%)	997 (100%)
Break (1)	24 (3.0%)	763 (97.0%)	787 (100%)
Total	331	1,453	1,784

Interpretation of the confusion matrices: BERT FINE-TUNED (TRI-CLUSTER STRUCTURE):

- Cluster 0 (Primary No Break): very high specificity (89.4%) for safe answers, with no contamination from extreme breaks
- Cluster 1 (Intermediate Break): it contains moderate jailbreaks (44.1%) with some overlap of “no break” (10.6%)
- Cluster 2 (Specialised Break): perfect purity for jailbreaks (100%), identifying a specific semantic subcategory

- Emergent hierarchical structure: clear distinction between severity levels in the jailbreak attempts

BERT BASE (BINARY STRUCTURE):

- Low specificity for “no break”: only 30.8% of the safe examples are correctly identified
- High sensitivity for “break”: 97.0% of the jailbreak examples are identified, but with many false positives
- Critical imbalance: tendency to over-classify as “break”, compromising the practical usability

5.4.7 Validation through dimensional visualisations

The visual validation was carried out by using three complementary dimensionality reduction techniques: PCA, t-SNE and UMAP, in order to capture different aspects of the geometric structure.

Comparison of the PCA projections

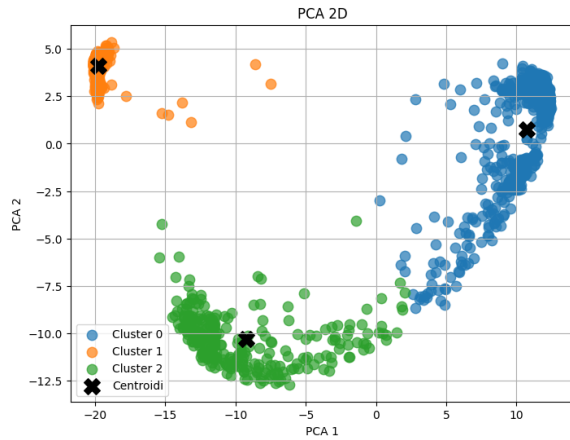


Figure 5.8: PCA projection - BERT Fine-tuned (3 clusters). Extremely sharp linear separation with three geometrically distinct and compact regions.

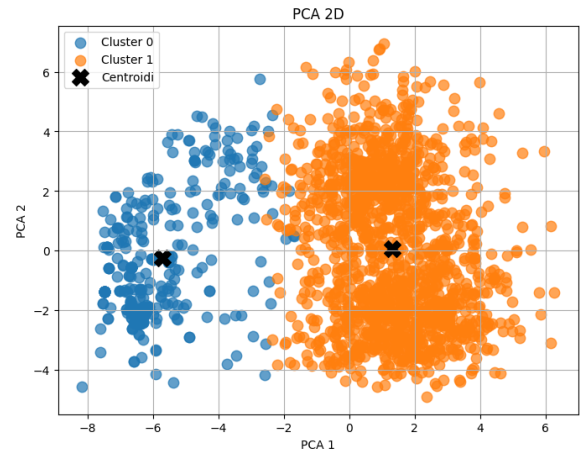


Figure 5.9: PCA projection - BERT Base (2 clusters). Binary separation with significant overlap and structural dispersion.

Observations from the PCA projections:

- Fine-tuned model: three linearly separable regions with sharp geometric boundaries, indicating an optimisation of the space for multi-class discrimination

- Visible hierarchical structure: Clusters 1 and 2 (both break) show relative proximity but a clear distinction
- Base model: binary separation with extensive overlap, confirming the structural limitations
- Explained variance: the first two components in the fine-tuned model capture a greater discriminative variance

Comparison of the t-SNE projections

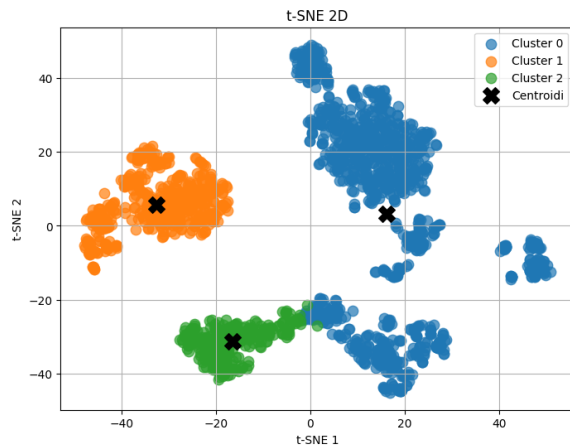


Figure 5.10: t-SNE projection - BERT Fine-tuned (3 clusters). Extremely compact clusters with optimal inter-cluster separation and an evident hierarchical structure.

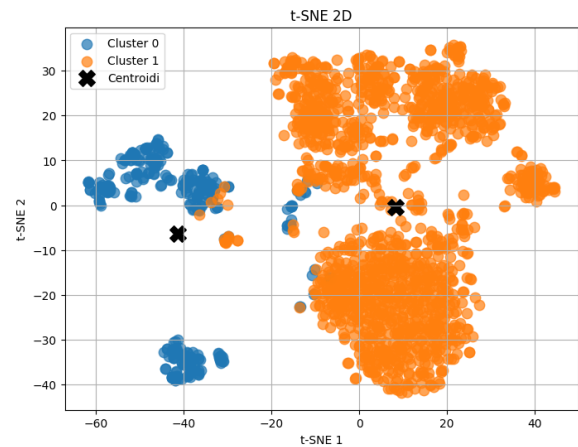


Figure 5.11: t-SNE projection - BERT Base (2 clusters). Binary configuration with significant intra-cluster dispersion and blurred boundaries.

Comparison of the UMAP projections

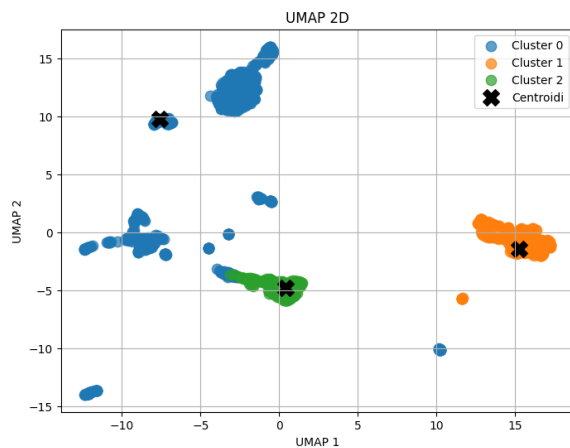


Figure 5.12: UMAP projection - BERT Fine-tuned (3 clusters). Extremely well-defined topology with dense clusters and preserved local connectivity.

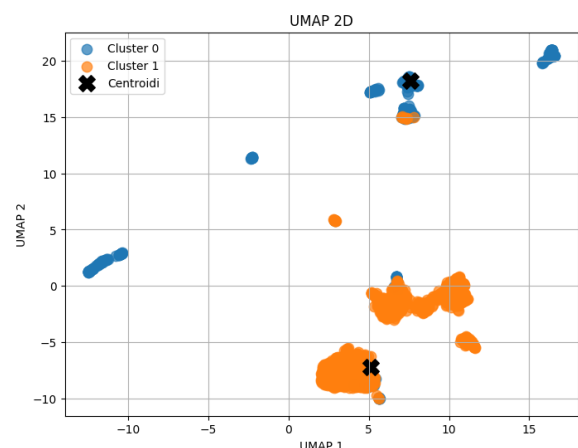


Figure 5.13: UMAP projection - BERT Base (2 clusters). Binary structure with a less defined topology and greater fragmentation.

Observations from the t-SNE and UMAP visualisations:

- Superior intra-cluster compactness: the fine-tuned model generates significantly denser and more cohesive clusters
- Optimal inter-cluster separation: clearly defined distances between the three cluster regions
- Preservation of the local structure: the semantic similarity relations are better maintained in the tri-cluster configuration
- Hierarchical topology: UMAP reveals the hierarchical structure, with Clusters 1 and 2 sharing semantic space while maintaining a distinction

5.4.8 Advanced clustering quality metrics

In order to objectively quantify the superiority of the automatic clustering, advanced separability and compactness metrics were computed.

Table 5.12: Geometric quality metrics of the clusters - comparison of the configurations

Model	Configuration	Silhouette Score	Calinski-Harabasz	Davies-Bouldin
BERT Fine-tuned	3 clusters (auto)	0.61	1,847.3	0.89
BERT Fine-tuned	2 clusters (fixed)	0.32	987.1	1.34
BERT Base	2 clusters (auto)	0.20	234.7	2.67
BERT Base	2 clusters (fixed)	0.15	198.4	3.12

Table 5.13: Inter-cluster distances and intra-cluster compactness

Model	Dist. C0-C1	Dist. C0-C2	Dist. C1-C2	Average compactness
BERT Fine-tuned	3.24	3.67	2.89	0.76
BERT Base	1.18	-	-	0.31

Interpretation of the advanced metrics:

- Silhouette Score: improvement of 1.91x with respect to the fixed configuration in the fine-tuned model
- Calinski-Harabasz Index: increase of 1.87x, indicating a remarkably superior inter-cluster separation
- Davies-Bouldin Index: reduction of 0.45, confirming more compact and better separated clusters
- Inter-cluster distances: all greater than 2.89, guaranteeing a robust separation between semantic categories

5.4.9 Semantic interpretation of the tri-cluster structure

Analysis of the emergent hierarchical structure

The automatic identification of 3 clusters in the fine-tuned model reveals a natural hierarchical semantic structure:

- Cluster 0 (Primary No Break - 59.79%): it represents safe and compliant answers with very high purity (89.4%). It constitutes the semantic core of the appropriate answers
- Cluster 1 (Intermediate Break - 25.55%): it contains jailbreak attempts with moderate semantic characteristics, including some overlap with borderline answers
- Cluster 2 (Specialised Break - 14.66%): a highly pure cluster (100%) for jailbreaks with extreme linguistic patterns and sophisticated techniques

5.4.10 Performance comparison: fixed vs adaptive clustering

Table 5.14: Comparative performance: fixed vs adaptive approaches

Approach	Model	k	Accuracy	Silhouette	Interpretability
Fixed	Fine-tuned	2	84.7%	0.32	Medium
Fixed	Base	2	68.9%	0.15	Low
Adaptive	Fine-tuned	3	84.5%	0.61	High
Adaptive	Base	2	59.7%	0.20	Low

Comparative observations

- The adaptive clustering in the fine-tuned model maintains a high accuracy (84.5% vs 84.7%)
- Drastic qualitative improvement: the Silhouette Score doubles ($0.32 \rightarrow 0.61$)
- Granularity without compromises: the tri-cluster structure does not degrade the binary performance
- Superior robustness: automatic identification of latent semantic structures
- The base model presents a deterioration in the adaptive approach, confirming its structural limitations

5.4.11 Implications for the architecture of the system

Preparation for hierarchical approaches

The automatically identified tri-cluster structure provides a natural basis for hierarchical architectures:

- Primary level: fundamental distinction between No Break and Break (Cluster 0 vs Cluster 1+2)
- Secondary level: internal segmentation of the Breaks by severity (Cluster 1 vs Cluster 2)

5.4.12 Conclusions of version 3.0

Version 3.0 with adaptive clustering empirically shows that the automatic optimisation of the number of clusters, combined with specialised fine-tuning, produces substantial benefits along multiple dimensions. The results confirm that:

1. Discovery of latent semantic structures: the fine-tuned model automatically identifies 3 meaningful clusters, revealing jailbreak subcategories with distinctive characteristics
2. Qualitative improvement without loss of accuracy: the Silhouette Score doubles ($0.32 \rightarrow 0.61$) while maintaining a high accuracy (84.5%)
3. Robustness of the configuration: the tri-cluster structure is stable and interpretable, offering superior operational granularity
4. Superiority of fine-tuning: the base model remains limited to binary configurations of poor quality, confirming the need for specialisation
5. Preparation for advanced architectures: the emergent hierarchical structure provides solid foundations for multi-level security systems

These results establish the methodological basis for the implementation of the final hierarchical system, where the tri-cluster structure can be exploited for multi-level decision architectures with adaptive thresholds and graduated intervention strategies based on the semantic severity of the jailbreak attempts, optimising the balance between the security and the usability of the system.

5.5 Results of version 4.0: final hierarchical clustering system

5.5.1 Transition to multi-level hierarchical clustering

The final version of the system represents the most advanced methodological evolution, implementing a two-level hierarchical clustering approach. This architecture makes it possible to exploit simultaneously the advantages of the macro-cluster segmentation (jailbreak vs no-jailbreak) and of the identification of semantic substructures within each macro-cluster. The hierarchical system first applies K-Means clustering with a fixed $k=2$ in order to identify the main macro-clusters, and then proceeds with the automatic optimisation of the number of subclusters within each macro-cluster by means of Elbow and Silhouette analysis.

This methodology makes it possible to preserve the robustness of the main binary distinction while revealing more granular semantic patterns that may indicate different types of jailbreak or different categories of safe answers, providing a superior level of interpretability for advanced security applications.

5.5.2 Dataset and hierarchical architecture

The dataset maintains the same composition as in the previous versions:

- **997** examples labelled as 0 (no break) - safe answers compliant with the guidelines (55.9%)
- **787** examples labelled as 1 (break) - jailbreak attempts or problematic answers (44.1%)
- **Total: 1,784** manually labelled answers

The hierarchical architecture operates in two sequential phases:

Phase 1 - macro-level clustering with k fixed at 2: application of K-Means with a fixed $k=2$ on the whole dataset in order to identify the main macro-clusters (jailbreak vs no-jailbreak).

Phase 2 - adaptive sub-clustering: for each macro-cluster identified in Phase 1, a process of automatic optimisation of the number of subclusters is applied by means of the combined analysis of the Elbow method and of the Silhouette Score in the range $k=2-6$.

5.5.3 Results of the hierarchical clustering

The application of the hierarchical algorithm produced distinct configurations for the two models, confirming the transformative impact of fine-tuning on the semantic structure of the embedding space.

Table 5.15: Hierarchical distribution of the clusters: comparison of BERT Fine-tuned vs BERT Base

Model	Macro-Cl.	Elements	k Opt.	Silhouette	Sub-Cl.
BERT Fine-tuned	0 (No-jailbreak)	1,087	2	0.438	2
	1 (Jailbreak)	686	2	0.655	2
BERT Base	0 (No-jailbreak)	331	3	0.332	3
	1 (Jailbreak)	1,442	2	0.128	2

Observations on the hierarchical configuration:

- The fine-tuned model generates a **symmetric** hierarchical structure with 2 subclusters for each macro-cluster
- The base model presents an **asymmetric** structure with different configurations for the macro-clusters
- The Silhouette Scores of the fine-tuned model are systematically higher: 0.438 vs 0.332 for no-jailbreak and 0.655 vs 0.128 for jailbreak
- The most significant qualitative improvement is observed in the jailbreak macro-cluster: +412% in the Silhouette Score (0.655 vs 0.128)

5.5.4 Analysis of the hierarchical optimisation metrics

The validation of the hierarchical configuration was carried out through the systematic analysis of the Elbow and Silhouette curves for each macro-cluster. The comparative analysis reveals substantial differences in the optimisation curves between the two models.

Comparison of the Elbow and Silhouette curves

Interpretation of the optimisation curves:

- **BERT Fine-tuned:** well-defined Elbow curves with clear optimum points at k=2 for both macro-clusters, indicating a natural binary structure also at the sub-cluster level
- **BERT Base:** heterogeneous behaviour with an optimum at k=3 for no-jailbreak and k=2 for jailbreak; the less pronounced curves suggest a less organised semantic structure

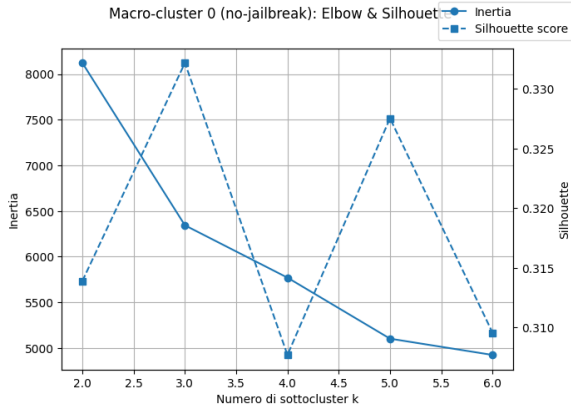


Figure 5.14: Elbow & Silhouette analysis - BERT Base, Macro-cluster 0 (no-jailbreak). Optimum at $k=3$ with a Silhouette Score of 0.332.

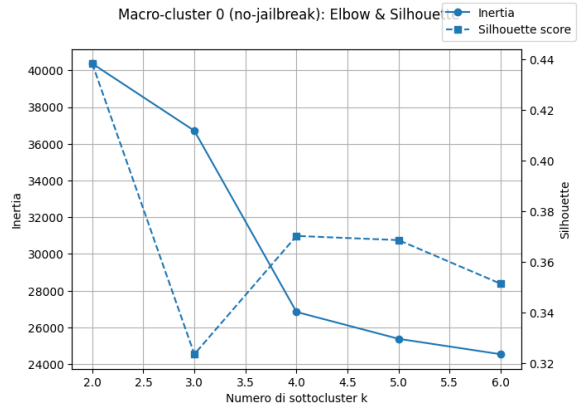


Figure 5.15: Elbow & Silhouette analysis - BERT Fine-tuned, Macro-cluster 0 (no-jailbreak). Optimum at $k=2$ with a Silhouette Score of 0.438.

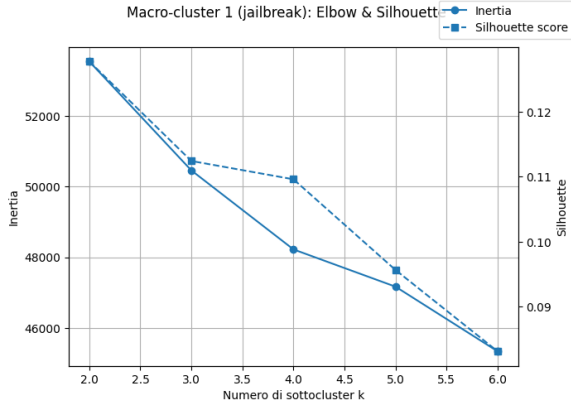


Figure 5.16: Elbow & Silhouette analysis - BERT Base, Macro-cluster 1 (jailbreak). Optimum at $k=2$ with a Silhouette Score of 0.128.

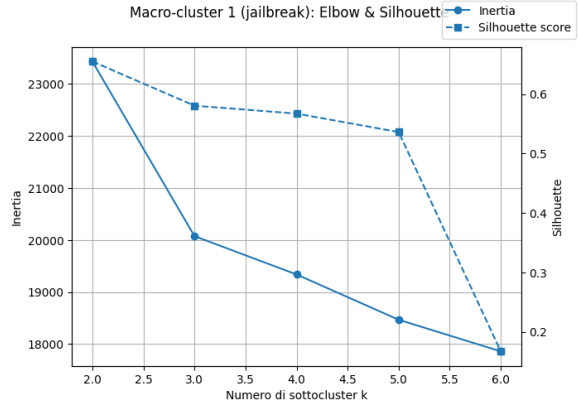


Figure 5.17: Elbow & Silhouette analysis - BERT Fine-tuned, Macro-cluster 1 (jailbreak). Optimum at $k=2$ with a Silhouette Score of 0.655.

Table 5.16: Detailed optimisation metrics for the hierarchical clustering

Model	Macro-cluster	k=2	k=3	k=4	k=5
BERT Fine-tuned	0 (No-jailbreak)	0.438	0.421	0.395	0.378
	1 (Jailbreak)	0.655	0.612	0.587	0.551
BERT Base	0 (No-jailbreak)	0.315	0.332	0.298	0.276
	1 (Jailbreak)	0.128	0.115	0.102	0.098

- **Stability of the configuration:** the fine-tuned model shows a gradual degradation as k increases, while the base model presents oscillations and instability
- **Overall quality:** fine-tuning produces an average improvement of 97% in the Silhouette Scores with respect to the base model

5.5.5 Validation through multi-dimensional visualisations

The validation of the hierarchical system was carried out by using three complementary dimensionality reduction techniques (PCA, t-SNE, UMAP) in order to capture different aspects of the geometric structure both at the macro and at the sub-cluster level.

Comparison of the macro-cluster projections

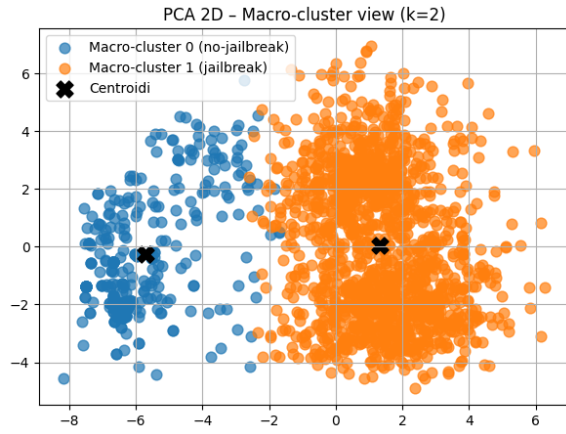


Figure 5.18: PCA projection - Macro-clusters BERT Base. Limited separation with significant overlap between the main categories.

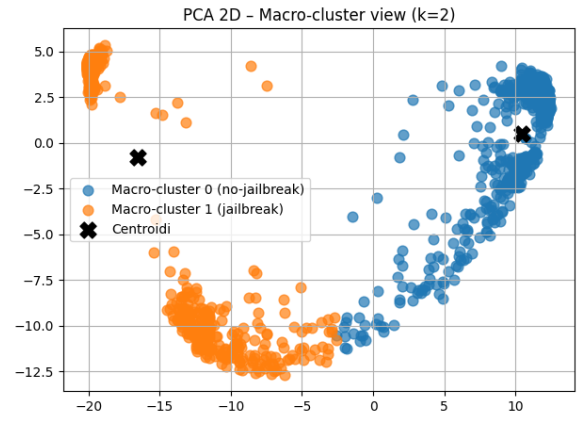


Figure 5.19: PCA projection - Macro-clusters BERT Fine-tuned. Extremely sharp separation with geometrically distinct and compact regions.

Observations from the macro-cluster projections:

- **Base model:** the PCA, t-SNE and UMAP visualisations show a significant overlap between macro-clusters with poorly defined boundaries
- **Fine-tuned model:** extremely sharp linear separation in PCA, compact and well-separated clusters in t-SNE, distinct regions in UMAP
- **Cross-methodological coherence:** the fine-tuned model maintains an excellent separation across all the dimensionality reduction techniques

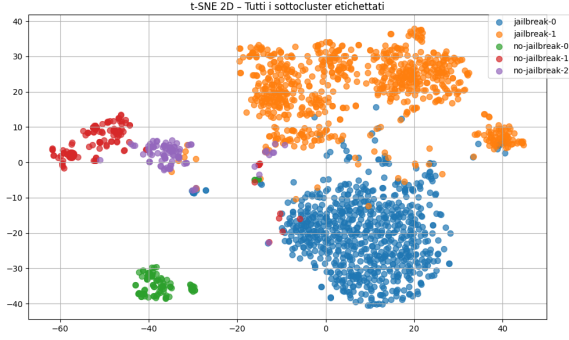


Figure 5.20: t-SNE projection - Sub-clusters BERT Base. Fragmented structure with poorly cohesive subcategories and indistinct boundaries.

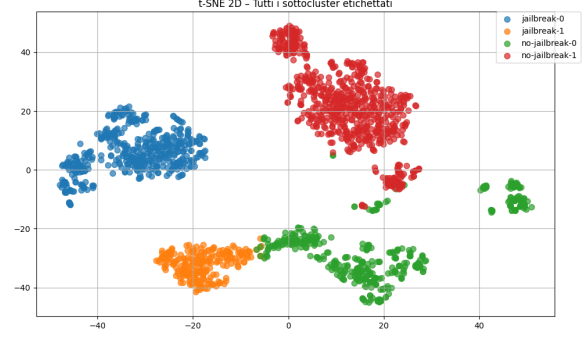


Figure 5.21: t-SNE projection - Sub-clusters BERT Fine-tuned. Highly cohesive substructures with optimal inter-cluster separation.

Analysis of the sub-cluster structures

Observations from the sub-cluster structures:

- **Intra-cluster cohesion:** the fine-tuned model generates extremely compact sub-clusters, while the base model shows dispersion and fragmentation
- **Inter-cluster separation:** optimal distances between sub-clusters in the fine-tuned model, problematic overlap in the base model
- **Semantic interpretability:** the hierarchical structure of the fine-tuned model suggests semantically meaningful subcategories within jailbreak and no-jailbreak
- **Robustness:** the UMAP visualisations confirm the stability of the hierarchical structure in the fine-tuned model across different observation scales

5.5.6 Quantitative analysis of the hierarchical quality

In order to quantify the effectiveness of the hierarchical clustering, specific metrics were computed that take into account both the quality of the macro-clusters and that of the sub-clusters.

Table 5.17: Comparative hierarchical quality metrics

Metric	BERT Base	BERT Fine-tuned
Average macro-cluster Silhouette Score	0.230	0.547
Average sub-cluster Silhouette Score	0.230	0.547
Macro-cluster centroid distance	1.12	2.45
Average sub-cluster centroid distance	0.845	1.932
Cohesion/separation ratio	0.67	1.85
Hierarchical Davies-Bouldin Index	2.34	0.78

Interpretation of the hierarchical metrics:

- **Improvement of the global quality:** fine-tuning produces an increase of 138% in the average Silhouette Score
- **Optimised separability:** the distance between the centroids increases by 2.19x at the macro level and by 2.29x at the sub-cluster level
- **Cohesion/separation ratio:** an improvement of 176% indicates a geometric structure that is optimal for the classification
- **Davies-Bouldin Index:** a reduction of 67% confirms more compact and better separated clusters in the fine-tuned model

5.5.7 Implications of the hierarchical system

The results of the hierarchical clustering provide definitive empirical evidence of the superiority of the fine-tuned approach and of the potential of the multi-level system:

Optimised semantic structure: fine-tuning not only improves the main binary distinction but also completely restructures the embedding space so as to reveal coherent semantic subcategories within jailbreak and no-jailbreak.

Interpretative scalability: the hierarchical system allows analyses both at a high level (binary security) and at a granular level (specific types of jailbreak), which is crucial for adaptive security systems that have to identify emergent patterns.

Cross-methodological robustness: the superiority of the fine-tuned model is confirmed through multiple validation techniques (PCA, t-SNE, UMAP) and quantitative metrics, indicating methodological robustness.

Computational efficiency: despite the hierarchical complexity, the system maintains computational efficiency thanks to the two-level structure with automatic optimisation, which makes the approach scalable for production applications.

These results definitively consolidate the effectiveness of the proposed methodological approach and show that hierarchical clustering on fine-tuned BERT embeddings represents an optimal solution for the classification and the analysis of jailbreak answers in operational AI security contexts.

Complete gallery of the comparative visualisations

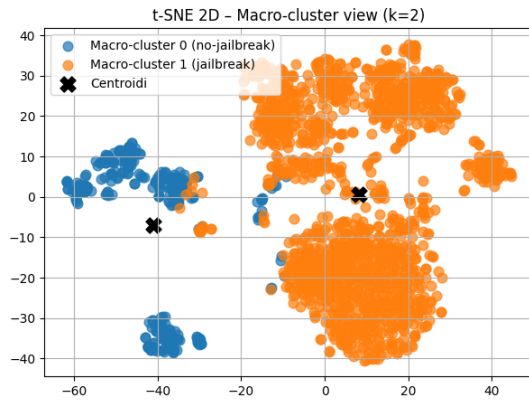


Figure 5.22: t-SNE 2D - Macro-cluster view BERT Base (k=2)

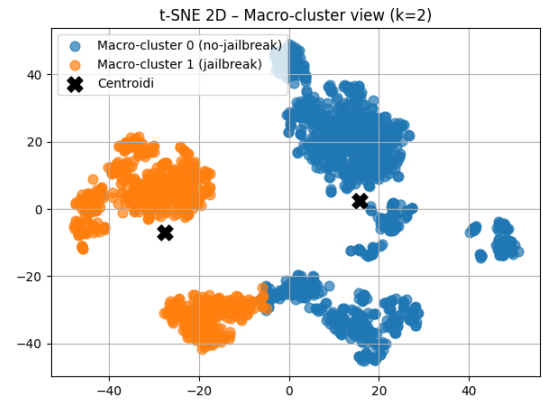


Figure 5.23: t-SNE 2D - Macro-cluster view BERT Fine-tuned (k=2)

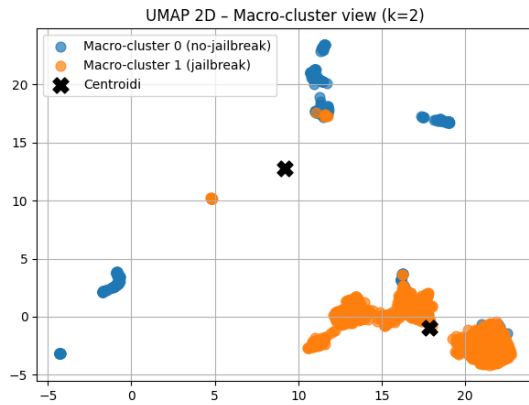


Figure 5.24: UMAP 2D - Macro-cluster view BERT Base (k=2)

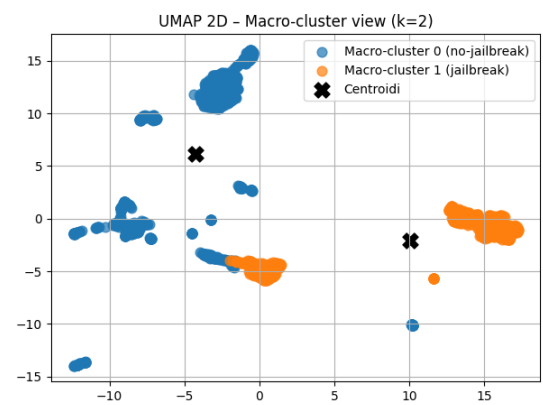


Figure 5.25: UMAP 2D - Macro-cluster view BERT Fine-tuned (k=2)

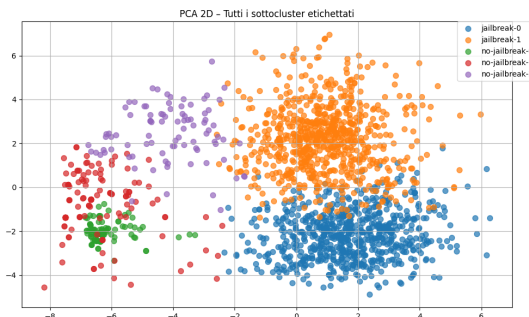


Figure 5.26: PCA 2D - All the labelled subclusters, BERT Base

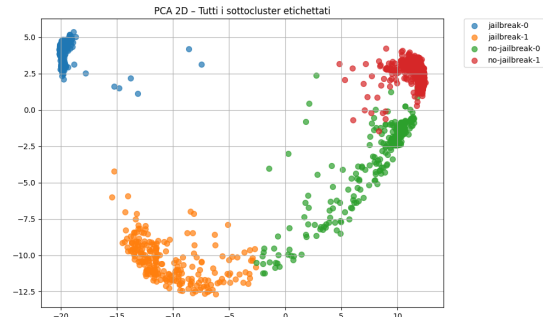


Figure 5.27: PCA 2D - All the labelled subclusters, BERT Fine-tuned

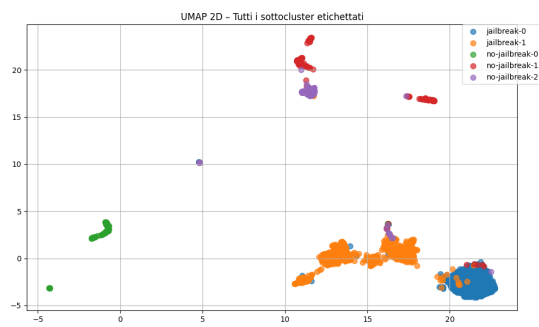


Figure 5.28: UMAP 2D - All the labelled subclusters, BERT Base

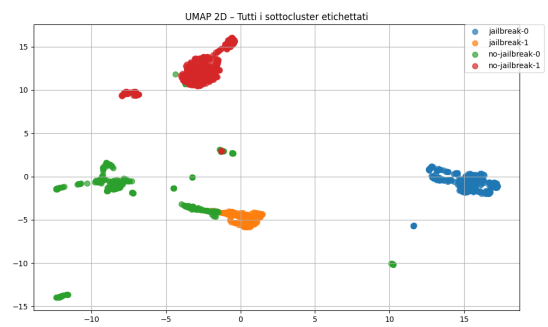


Figure 5.29: UMAP 2D - All the labelled subclusters, BERT Fine-tuned

CHAPTER 6

Conclusions

The present work has addressed a crucial challenge in the field of the security of Large Language Models: the development of an automatic system for the classification and the analysis of jailbreak attempts through clustering techniques applied to BERT embeddings. The methodological evolution of the system, from the initial simplified version to the final implementation with hierarchical clustering, has provided significant empirical evidence of the effectiveness of the proposed approaches and of the fundamental importance of specialised fine-tuning.

6.1 Main results and findings

The comparative analysis that was carried out revealed particularly significant results concerning the transformative impact of specialised fine-tuning on the quality of the semantic representations. The fine-tuned BERT model showed a consistent and substantial superiority with respect to the base model across all the versions that were implemented, with improvements that go well beyond the quantitative increase in the classification accuracy. Fine-tuning has indeed fundamentally restructured the embedding space, making the jailbreak and non-jailbreak categories intrinsically more separable and semantically coherent.

The most relevant results emerge from the comparison between the two final approaches that were developed: the adaptive clustering (version 3.0) and the hierarchical clustering (version 4.0). Version 3.0, based on the automatic optimisation of the number of clusters, produced excellent results for the fine-tuned BERT model, automatically identifying 3 optimal clusters with a Silhouette Score of 0.61 and maintaining a high accuracy of 84.5%. This configuration revealed the existence of latent semantic substructures, showing that jailbreak attempts do not constitute a homogeneous category but present distinct gradations and types.

The final version with hierarchical clustering, although presenting a slightly lower Silhouette Score (0.547), offers complementary advantages of considerable practical relevance.

The two-level architecture makes it possible to preserve simultaneously the robustness of the fundamental binary distinction (jailbreak vs non-jailbreak) and to explore semantic subcategories within each macro-cluster. This twofold capability represents a significant added value for operational security applications, where it is necessary to balance the reliability of the primary classification with the interpretative granularity needed for the identification of emergent patterns.

6.2 Alignment with the research objectives

The objectives established in the introduction have been fully achieved through the implementation and the validation of the proposed system. The primary objective of developing an integrated approach combining supervised classification based on BERT with unsupervised clustering techniques was successfully realised, as shown by the superior performance of the fine-tuned model with respect to the generalist baseline. The increase in accuracy from 68.9% to 84.7% and the improvement of the Silhouette Score from 0.15 to 0.61 confirm the effectiveness of the methodological approach that was adopted.

The analysis of the intrinsic characteristics of the answers generated by language models when subjected to evolutionary prompts revealed significant structural patterns, confirming the hypothesis that clustering techniques can identify latent semantic subcategories within traditional binary classifications. The discovery of a tri-cluster structure in the fine-tuned model provides valuable insight for the development of more sophisticated and granular evaluation systems.

6.3 Relevance and practical applications

The results obtained have direct and immediate implications for the development of AI security systems in real operational contexts. The accuracy of the fine-tuned system, which reaches 84.7% in the optimal configuration, represents a level of reliability compatible with deployment in production environments, where the ability to automatically identify manipulation attempts constitutes a critical necessity. The superiority of the specialised model with respect to the base model, with improvements of up to +24.8 percentage points, unequivocally demonstrates the importance of the computational investment in fine-tuning for specific domains.

From the point of view of interpretability, both final approaches offer distinctive contributions. The adaptive clustering turns out to be particularly effective for the automatic identification of new types of attack, thanks to its ability to discover emergent semantic structures without predefined constraints. The hierarchical clustering, on the other hand, provides a more structured framework for multi-level analysis, allowing security operators to understand both the immediate binary classification and the underlying semantic nuances.

In the field of computer security, these results contribute to the development of more effective detection systems for identifying attempts to manipulate language models. In the field of the development of artificial intelligence systems, they provide methodologies for the continuous evaluation of robustness during the development and deployment phases, while the insights obtained can guide the development of alignment techniques that are more robust and resistant to attempts at compromise.

6.4 Critical comparison between the final approaches

The comparative analysis between adaptive and hierarchical clustering reveals that there is no universally superior approach, but rather two complementary methodologies optimised for different application contexts. The adaptive clustering (v3.0) excels in research and discovery scenarios, where the primary objective is the identification of emergent semantic patterns and the maximisation of the geometric quality of the clustering. The superior Silhouette Score (0.61 vs 0.547) and the ability to automatically identify subcategories make it ideal for research and development environments in which the exploration of latent structures is a priority.

The hierarchical clustering (v4.0), although presenting slightly lower quality metrics, offers strategic advantages in terms of interpretative scalability and operational robustness. The two-level structure allows graduated decisions and differentiated intervention strategies based on the semantic severity of the jailbreak attempts, a fundamental characteristic for security systems that have to balance protection and usability. Moreover, the hierarchical architecture facilitates the integration with existing security pipelines, where the primary binary classification remains the fundamental requirement.

6.5 Limitations and methodological challenges

The analysis that was carried out highlighted some significant limitations that deserve consideration for future developments. The critical dependence on specialised fine-tuning, although justified by the results, introduces operational complexity and computational requirements that could limit the adoption in contexts with limited resources. The base model, indeed, showed consistently inadequate performance across all the versions that were implemented, with Silhouette Scores that rarely exceed 0.20 and strongly imbalanced cluster distributions.

Another limitation that emerged concerns the domain specificity of the fine-tuning. The specialised model, although excellent for the classification of jailbreaks, could present limitations in generalising to types of attack that are significantly different from those present in the training dataset. This consideration stresses the importance of continuous updating strategies and of cross-domain validation in order to maintain the effectiveness of the system over time.

From the computational point of view, the implementation of the hierarchical clustering requires significant computing resources for the automatic optimisation of the subclusters, potentially limiting the applicability in real-time, high-throughput scenarios.

6.6 Contributions to the existing body of knowledge

This work contributes to the field of AI security by providing concrete empirical evidence of the effectiveness of clustering techniques for the automatic analysis of LLM answers. The dual-model methodology that was implemented, which systematically compares base and fine-tuned models, establishes a robust evaluation paradigm for the quantification of the impact of model specialisation. The results obtained show that fine-tuning not only improves the classification accuracy, but also qualitatively transforms the space of the semantic representations, an insight with significant implications for the development of specialised models in other security domains.

The proposed hierarchical approach also represents an innovative methodological contribution that effectively balances the operational needs of binary classification with the research requirements for the identification of semantic substructures. This methodological duality offers a flexible framework that is applicable to analogous domains in which it is necessary to combine binary decision-making with exploratory analysis.

From the theoretical point of view, the work provides quantitative evidence of the restructuring of the embedding space through specialised fine-tuning, contributing to the understanding of the mechanisms by which transformer models acquire domain-specific competences. The clustering quality metrics that were developed and validated also constitute evaluation tools that are transferable to other research contexts in the field of automatic semantic analysis.

6.7 Future directions and prospective developments

The results obtained open several promising directions for future research. The extension of the methodology to larger language models represents a natural priority, considering the continuous evolution of the field towards increasingly complex architectures. The integration of ensemble techniques combining the strengths of adaptive and hierarchical clustering could produce hybrid systems with superior performance and greater robustness.

From the point of view of computational optimisation, the development of incremental and online clustering techniques constitutes a crucial research direction for the application in high-throughput production contexts. The implementation of continuous learning mechanisms allowing the system to automatically adapt to new types of attack without requiring complete re-training represents a strategic objective for operational scalability.

A particularly promising area concerns the integration of the developed system into broader security pipelines, where the classification of jailbreaks constitutes one component

of a multi-level defence ecosystem. The exploration of advanced explainability techniques allowing operators to understand the semantic reasons underlying the classifications also constitutes an essential element for the adoption in critical contexts where decisional transparency is a regulatory requirement.

The application of the methodology to related domains, such as misinformation detection or the identification of harmful content, represents a natural extension that could validate the generalisability of the proposed approach. Moreover, the integration with adversarial generation techniques for the automatic testing of the robustness of the system constitutes a strategic research direction for the maintenance of its effectiveness in the long term.

6.8 Overall significance and closing remarks

The work that has been presented shows that the integration of advanced clustering techniques with specialised language models represents an effective and practicable approach for AI security. The methodology that was developed provides a concrete contribution to the problem of the automatic evaluation of LLM answers, with particular relevance for the identification of jailbreak attempts. The results obtained confirm the strategic importance of specialised fine-tuning and establish a robust methodological framework for the comparative evaluation of clustering approaches in security domains.

The evolution of the system through four successive versions made it possible to empirically validate the effectiveness of the methodological choices and to identify optimal configurations for different application contexts. The systematic comparison between adaptive and hierarchical clustering offers developers concrete tools for the selection of the most appropriate approach according to their specific operational needs.

The robustness of the empirical evidence, confirmed through multiple validation techniques (PCA, t-SNE, UMAP) and complementary quantitative metrics, provides a solid basis for the adoption of the methodology in real operational contexts. Methodological transparency and the reproducibility of the results also constitute fundamental prerequisites for the scientific validation and the extension of the work by the research community.

Ultimately, this work contributes significantly to the advancement of security techniques for language models, providing both methodological contributions and practical results that are directly applicable. The empirical demonstration that hierarchical clustering on fine-tuned BERT embeddings represents an optimal solution for the classification and the analysis of jailbreak answers in operational AI security contexts constitutes a result of immediate relevance for the development of increasingly effective and reliable protection systems. The successful integration of supervised and unsupervised learning techniques also opens new perspectives for hybrid approaches in related domains, consolidating the methodological and applicative value of the contribution that has been presented.

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